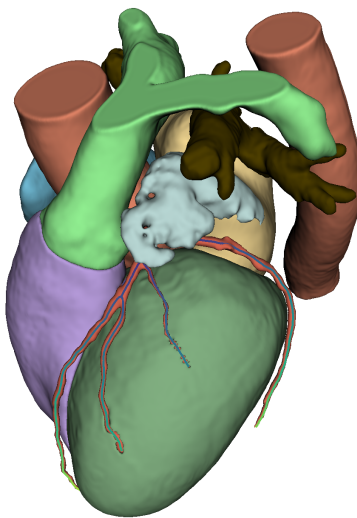


Tracing the coronary arteries in 3D computed tomography scans

Master Thesis



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Cover photo: 3D rendering of a heart volume with heart structures from TotalSegmentator [1], coronary arteries segmentation from ImageCAS [2], and extracted centerlines.

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Abstract

Cardiovascular diseases (CVDs) are a leading cause of global deaths. One of the most serious forms of CVDs is coronary artery disease (CAD). It is caused by atherosclerosis, characterized by plaque building up in the arteries, resulting in a disrupted blood flow. Coronary computed tomography angiography (CCTA) is a typical, non-invasive method of diagnosing CAD. Automatic extraction of coronary artery centerline from CCTA scan is vital for effective CAD diagnostic process. Moreover, it is a fundamental step for further coronary artery research, including lumen analysis and stenoses detection. This project proposes a coronary tree tracing algorithm, that requires only a CCTA scan and a location of the beginning of the coronary tree to trace the whole left coronary artery (LCA) structure.

The proposed solution is based on four models: centerline tracer, branch point detector, artery radius estimator, and artery likelihood estimator. All models utilize convolutional neural networks (CNNs). The centerline tracer estimates the direction in which the tracer should follow to mimic the blood flow. Branch point detector is used for locating branch points based on likelihood. Radius estimator helps approximate the radius of coronary artery, given a point on the centerline. The artery likelihood estimator outputs a likelihood of a given point being inside LCA. The entire tracing algorithm is based on alternating centerline tracing and detecting branch points on it. Artery likelihood estimator is used as a robust tool for terminating the tracing and, together with radius estimator, is incorporated in branch spawning method indicating the directions of subsequent branches.

The algorithm was trained and evaluated on a curated version of publicly available ImageCAS dataset. All four models perform exceptionally. Centerline tracer reaches an average weighted accuracy of 0.87 in predicting centerline directions. Branch detector has a recall of 0.84. Radius estimator successfully approximates radii of LCA with an average RMSE of 0.17. Artery likelihood estimator achieved AUC score equal to 0.96, proving itself a useful tool for terminating the tracing and branch spawning. The full coronary tree tracing algorithm reached an overlap of 81.2, being comparable to state of the art methods, and average inside of 0.097 mm, surpassing other available methods. The algorithm's generalization performance was tested on 939 CCTA scans from Copenhagen General Population Study (CGPS), showing its ability to capture the vast majority of LCA structures.

A secondary goal of this project was to detect phenotypes of the left main coronary artery (LMCA). Such a solution could assist in identifying high risk patients and gaining insight into causes of plaque formation. Three different phenotype profiles were derived through hierarchical clustering, with their characteristics successfully linked to known risks of plaque development.

Preface

This thesis proposes a method for tracing coronary artery centerline in 3D computed tomography scans.

The thesis was prepared in collaboration with the Section for Visual Computing at DTU Compute, the Department of Applied Mathematics and Computer Science, and the Department of Cardiology at Rigshospitalet, University of Copenhagen.

The project lasted from February 3rd 2025 to August 3rd 2025 and was carried out in fulfilment of the requirements for Master of Science degrees in Human-Centered Artificial Intelligence and Business Analytics at the Technical University of Denmark (DTU).

The reader is assumed to have a fundamental knowledge of machine learning, deep learning, and image analysis concepts.



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Last but not least, we want to thank our families and friends for their invaluable support that we received both throughout the project and the whole course of studying.

Abbreviations

AI - Average Inside
AIC - Akaike Information Criterion
AUC - Area Under the Curve
BIC - Bayesian Information Criterion
CAD - Coronary Artery Disease
CCA - Conventional Coronary Angiography
CCHS - Copenhagen City Heart Study
CCTA - Coronary Computed Tomography Angiography
CGPS - Copenhagen General Population Study
CNN - Convolutional Neural Network
CPR - Curved Planar Reformatting
CT - Computed Tomography
CVD - Cardiovascular Disease
DDQN - Double Deep Q-Network
FN - False Negative
FP - False Positive
HU - Hounsfield Units
kNN - k-Nearest Neighbours
LA - Left Atrium
LAD - Left Anterior Descending
LCA - Left Coronary Artery
LCX - Left Circumflex Artery
LM - Left Main
LMCA - Left Main Coronary Artery
LV - Left Ventricle
NN - Neural Network
OV - Overlap
PDA - Posterior Descending Artery
RA - Right Atrium
RCA - Right Coronary Artery
ReLU - Rectified Linear Unit
RI - Ramus Intermedius
RL - Reinforcement Learning
RMA - Right Marginal Artery
RMSE - Root Mean Square Error
ROC - Receiver Operating Characteristic
RV - Right Ventricle
TPM - True Positive Method
TPR - True Positive Reference

vMF - von Mises-Fisher

VMTK - The Vascular Modelling Toolkit

VTK - The Visualization Toolkit

WHO - World Health Organization

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Cardiovascular diseases (CVDs) are the most common global cause of deaths. In 2021 around 20.5 million people died due to CVD accounting for around 1/3 of global deaths. About half of them (49.2%) was caused by heart attacks while strokes contributed to around 25% [3]. Both of them are usually caused by disruption in blood flowing to the heart or brain. It is estimated that CVD affects over 500 million people globally.

Disease of blood vessels responsible for supplying the heart muscle, called a coronary heart disease, is directly linked to coronary artery which is the major blood vessel pumping the blood into the heart. Patients suffering from it have a disrupted blood flow caused by accumulated build-ups of fatty deposits on the inner walls of the coronary artery. Diagnostics of coronary artery disease (CAD) is therefore crucial to guide appropriate treatment in order to prevent heart failures [4, 5].

One way to diagnose CAD is through a coronary computed tomography angiography (CCTA). This modality has been proven to be effective and since it is a non-invasive procedure, it has been widely recommended. The problem with it, however, is that mapping of the coronary artery based on the 3D image volume is usually done manually by the experts which makes it time-consuming and error-prone [6].

The main purpose of this project is to propose a solution for automated tracing of the coronary artery centerline based on CCTA scans. Our solution combines several deep learning models aiming at finding tracing directions, detecting bifurcation points, estimating radius of the artery and estimating likelihood of being inside an artery, into one final tracing algorithm. The method proposed by us requires only 3D image volume and a root point of the coronary artery to recreate the centerline of the whole tree.

The tracing method can act as a complementary research tool that, together with methods used for analysing lumen or plaque composition, can create a framework for understanding the risk factors in CVDs [5, 7, 8]. Another potential use case is assisting with the time-intensive task of annotating coronary arteries, typically performed by medical imaging experts [6]. Our proposed approach has the potential to significantly accelerate this process. Moreover, when combined with complementary techniques for reconstructing the arterial structure around the centerline, our algorithm could enable full automation of the annotation workflow.

As a secondary objective, our project looks into the characteristics behind coronary artery trees' anatomy to derive several phenotypes. Focusing at left main coronary artery (LMCA), we extract a collection of measurements regarding artery length, radius and branching angles, and apply a clustering method to uncover phenotype profiles. Distinctive features of the phenotypes are linked to risk factors of cardiovascular diseases. The phenotype analysis, although here performed on ground truth

data, may as well be used along with the tracing algorithm making it deployable on a larger scale. The solution can be used to identify patients at increased risk, either in clinical field or as a tool to support research focused on CAD.

CHAPTER 2

Literature Review

Automatic coronary tree extraction has been a subject of research for the last few decades. One of the first attempts [9, 10] involved minimal cost path methods, where a path between two or more points is found by minimizing the cost function. The cost between given locations is the lowest when they both lie on a centerline. Those methods, however, struggle with tortuosity and often tend to make shortcuts, meaning that they can potentially produce results which do not match the anatomical structure.

More novel approaches include iterative tracking combined with convolutional neural networks (CNNs). Wolterink et al. [11] proposed a method using a 3D dilated convolutional neural network for predicting most plausible direction of an artery together with its radius in a CCTA image. Once the direction is established, a seed is placed in an artery either in a manual or automatic way, and a centerline is tracked further. While this method achieves state-of-the-art results for tracking individual coronary arteries, tracking the whole coronary tree requires many seed points out of which many overlap with centerlines extracted before, making the whole process time-consuming and computationally ineffective.

An extension of approach presented by Wolterink et al. was proposed by Leccardi et al. [12]. They introduced a multi-path deep CNN that estimates the vessel radius, and moving directions at each location. What is more, they expanded the iterative tracker using seed spawning, and introduced additional post-processing stage using a graph-based method. Combining all of those methods, they managed to track the whole coronary tree using only two initial seed points: one for left, and one for the right arterial branch. The downside of this approach is that its explorative nature requires spreading of many seeds around the traced path affecting the tracing computational time.

Another noteworthy approach was proposed by Radutoiu [13]. Based on convolutional neural networks, the method combines a tracer tasked to follow the centerline with a branch detector which is responsible for detecting bifurcation points along the traced centerline. The study also introduces a solution for detecting the starting point of the coronary artery tree. Unlike many prior studies in the literature, the algorithm was trained on a publicly available ImageCAS dataset [2] of hundreds of scans. The work resulted in a robust algorithm able to detect the coronary artery tree fully automatically in a computationally efficient manner.

Zhang et al. [14] proposed a branch-aware coronary centerline extraction method (BACCE) based on Double Deep Q-Network (DDQN) and 3D dilated CNN. DDQN is responsible for coronary centerline tracking, whereas CNN detects if a given point in the image is a branch point and calculates the radius. The big advantage of this method is that only one seed is needed for the entire coronary tree extraction. It also

handles tracing of coronary arteries that suffer from plaque or stenosis well.

A reinforcement learning (RL) approach enabling extracting the whole tree structure with only one initial point was proposed by Li et al. [15]. They presented a deep reinforced tree-traversal agent - a RL-based method with sequential decision-making. They also introduced a multi-task discriminator detecting bifurcations and vessel terminations. The method yielded reliable results on a dataset of considerable size.

Another worth mentioning approach focuses on generalizing algorithm's tracking ability to other vessels within the human body. SIRE proposed by Albas et al. [16] concentrates on scale-invariant and rotation-equivariant estimation of arteries orientation. The approach is aimed at solving two issues faced by most state-of-the-art solutions, one being lack of generalization to different types of blood vessels, and the other being the varying vessel tortuosity between patients. Authors propose a graph neural network utilizing multiple, parallelly operating, nested spherical meshes with different sizes. The method has been tested on several different training datasets containing various vessels. The results showed potential in utilizing the generalizing abilities of the model to automate various medical image processes.

Paper	Methods	Test scale	Overlap	Average Inside
Radutoiu	CNN	86	62.5	NA
Wolterink et al.	CNN	24	93.7	0.21
Leccardi et al.	Multi-path CNN	8	LAD = 100.0 LCX = 98.6	LAD = 0.26 LCX = 0.36
Zhang et al.	DDQN, CNN	8	96.2	0.21
Li et al.	RL	140	90.1	0.25*
Albas et al.	Graph NN	8	85	0.35
Wink et al.	Minimum cost path	32	NA	0.65*
Krissian et al.	Minimum cost path	16	90.9	0.39

Table 2.1: Comparison of most prominent coronary artery tracing methods. Average inside expressed in millimetres * - slightly modified versions of AI.

Shifting focus to phenotypes, an interesting work has recently been published by Wang et al. [17]. The study performed an unsupervised clustering to uncover phenotypes based on morphology of left main coronary artery (LMCA) fragments of 76 patients. Further on, the authors applied methods for evaluating the fluid dynamics within arteries of each phenotype. The phenotypes' characteristics were successfully linked to health risks indicating potential in detecting patients with risk of developing coronary atherosclerosis.

CHAPTER 3

Clinical Background

3.1 Heart anatomy

Heart is an organ that is responsible for supplying blood to the body. It consists of four chambers: **right atrium (RA)**, **left atrium (LA)**, **right ventricle (RV)** and **left ventricle (LV)**. The atria are smaller than the ventricles. RA sends the deoxygenated blood into the RV after receiving it from **vena cava** - human body's largest vein. RV then pumps the blood into the pulmonary circuit, where it is oxygenated. Afterwards, oxygenated blood comes back into the LA and is then sent into LV. LV then pumps the blood into the rest of the body. At the end, the blood returns through vena cava to the right atrium and the whole cycle repeats. [18, 19] The circulatory system of a human is visible in figure 3.1.

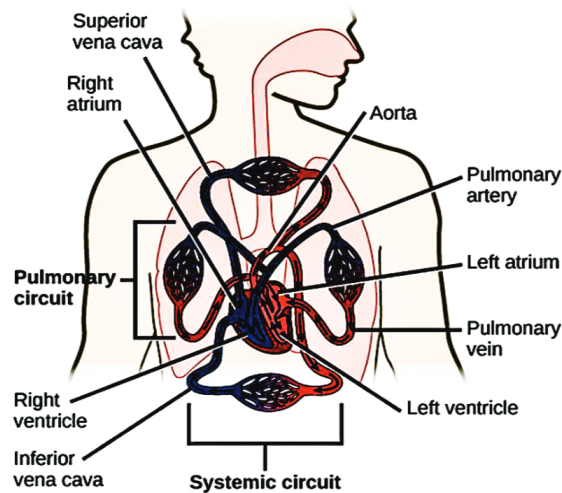


Figure 3.1: Circulatory system of a human, comprising systemic, pulmonary, and coronary circuits. Blood enters right atrium from systemic circuit through vena cava, is sent to right ventricle and then to pulmonary circuit. Oxygenated blood comes back to cardiac circuit: first to left atrium, then to left ventricle. Finally it exits the cardiac circuit through the aorta and is sent to other organs. Image from [20].

The cardiac cycle in which heart receives blood and then ejects it through the arteries

is divided into diastolic phase (diastole), in which heart chambers are relaxed receiving blood, and systolic phase (systole), when the blood is sent out through chambers' contractions. The atria and ventricles take turns contracting and relaxing. When the atria are in diastolic phase, the ventricles are in systolic phase, and vice versa [21]. There are four valves in the heart that ensure synchronization between those phases and the correct direction of the passage of blood by appropriate opening and closing. There are two types of valves: atrioventricular valves, including the tricuspid and mitral valves, and semilunar valves, including the pulmonary and aortic valves. Atrioventricular valves connect atria with ventricles: tricuspid one on the right side of the heart, and mitral one on the left side. Semilunar valves are responsible for blood flow between ventricles and arteries: pulmonary valve is connected to the right ventricle, and aortic valve to the left ventricle [18, 19, 22]. Passage of the blood throughout heart's chambers and valves is demonstrated in figure 3.2.

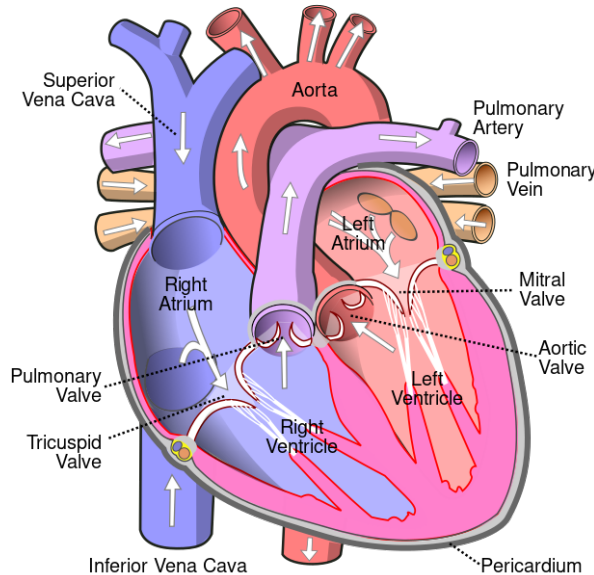


Figure 3.2: The passage of blood in the human's heart. Image from [23].

Cardiac muscle, called myocardium, is responsible for contraction and relaxation of the heart, enabling it to receive and pump blood. Together with two other muscle tissues it forms the heart wall, with myocardium being the thickest layer. Coronary arteries supply blood to the cardiac muscle, facilitating its contractile function [18].

Coronary arteries encircle the heart, with small branches extending into the myocardium to supply it with blood. There are two main coronary arteries: **left coronary artery (LCA)** and **right coronary artery (RCA)**. They branch from the

base of the aorta (human body's largest artery). Specifically, right and left coronary artery arise from right and left cardiac sinuses, respectively, that are two out of three dilations in aorta's wall just at the base of the aorta [18, 19]. Coronary arteries are shown in the figure 3.3

Left main coronary artery (LM or LMCA) is a short vessel, that branches into **left anterior descending artery (LAD)** and **left circumflex artery (LCX)**. In some cases the LMCA might not be present with LAD and LCX emerging directly from the left cardiac sinus. LAD, together with its branches, supplies both the ventricles and interventricular septum (a wall separating ventricles). LAD mainly branches into a varied number of **diagonal** branches. LCX primarily supplies the left atrium and parts of the left ventricle. It branches into one or more (**obtuse**) **marginal** arteries [18, 19, 24]. In around 15% of the population, LMCA trifurcates, giving rise to LAD, LCX, and **ramus intermedius (RI)** [25]. Studies show that RI, when present, has smaller length than LAD and LCX [26].

The RCA major branches include the **posterior descending artery (PDA)**, supplying interventricular septum, and the **right marginal artery (RMA)**, that provides blood to the right ventricle [18, 19, 24].

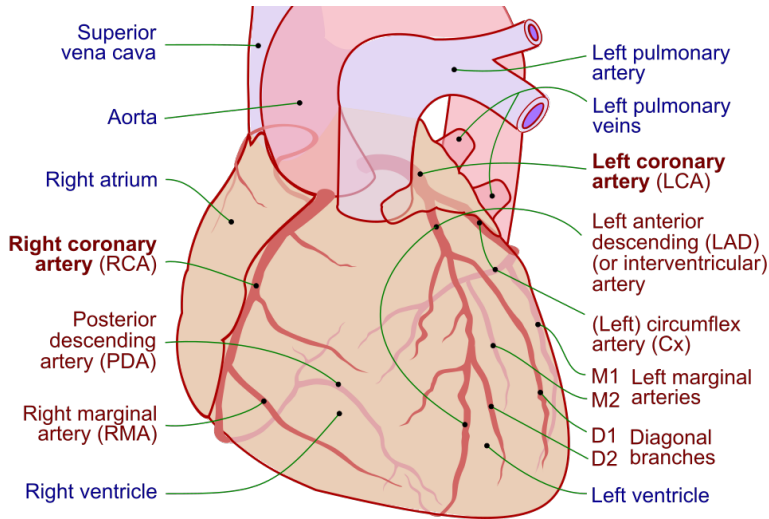


Figure 3.3: Coronary arteries (red text) and other key landmarks (blue text). Image from [27].

Depending on where the PDA branches of, we define three types of heart dominance [19, 24]:

- right-heart dominance, when PDA originates from RCA [19, 24], observed in

around 70% of the population [28],

- left-heart dominance, with PDA branching of LCX [19, 24], observed in 5-10% of the population [28],
- co-dominance, when PDA is a result of anastomosis created between LCX and RCA [19, 24], observed in the rest of the population [28].

3.2 Coronary artery disease (CAD)

Cardiovascular diseases (CVD) affect over 500 million people globally and are the primary cause of deaths worldwide [3]. In 2021, cardiovascular diseases, including heart disease, brain-related vascular diseases, and blood vessel disorders, were responsible for 20.5 million deaths [29]. The death toll is projected to rise to 35.6 million by 2050 [30]. CVDs can be classified in the following way [31]:

1. Caused by atherosclerosis:

- Ischemic or coronary artery disease (CAD) (e.g. heart attack),
- Cerebrovascular disease (related to the brain – e.g. stroke),
- Diseases of the aorta and arteries (e.g. hypertension or peripheral vascular disease).

2. Other:

- Congenital heart disease,
- Rheumatic heart disease,
- Cardiomyopathies,
- Cardiac arrhythmias.

The most serious across the population are by far the first two caused by atherosclerosis - CAD and cerebrovascular disease accounting for about 75% of CVD related deaths [3]. Some studies indicate that in United States more than half of the 45- to 84-year-olds can suffer from atherosclerotic disease and may not be aware of this [32].

Atherosclerosis is a condition in which, over time, inner walls of arteries accumulate fat, cholesterol, calcium and other substances [33, 34]. The build-up deposits, called plaques, can narrow down the lumen limiting the blood flow. In the most severe scenario plaques can rupture and form a blood clot which obstructs the circulation. When atherosclerosis occurs in the coronary artery it is called a coronary artery disease. As coronary arteries are responsible for supplying the heart muscle with blood together with oxygen and nutrients, their blockage can cause serious consequences. Blocked blood flow leads to myocardium being deprived of oxygen which damages it

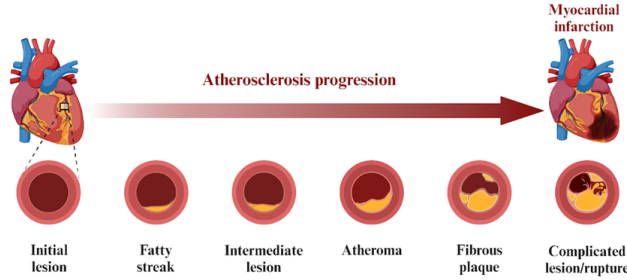


Figure 3.4: Atherosclerosis development. Image from [35].

causing, often fatal, myocardial infarction (heart attack). The process of atherosclerosis development is shown in figure 3.4.

Atherosclerotic disease is a slowly developing disease which often does not show any symptoms until it reaches its late stages. By that time, it might be too late to prevent its serious consequences. This highlights the importance of early detection during the subclinical phase to maintain a healthy population [4, 5].

One of the most serious forms of artery blockages are those happening in the left main coronary artery. CAD located in LMCA is associated with high mortality and poor clinical outcomes [36]. According to research, certain geometric parameters of the LMCA are linked to the presence of atherosclerotic plaques. One of such features is LMCA length: research found it was shorter in patients with coronary artery disease. Short LMs were also related to the stenoses location associated with higher severity [37]. High tortuosity and curvature of arteries were associated with plaques formation [38]. Another study suggested negative correlation between the LM branch angle and atherosclerotic disease formation in daughter branches (LAD and LCX) [39]. All of those geometric features and their association with the coronary artery disease are strictly related to the coronary artery hemodynamics and blood flow patterns.

Different patterns of wall shear stress can have different effects on plaque development. Low shear stress is linked to a higher likelihood of plaque formation, whereas elevated shear stress is associated with an increased risk of rupture [40]. Insights into patient's coronary arteries geometry could potentially help determine risk of developing atherosclerotic disease and guide the implementation of prevention measures.

3.3 Coronary computed tomography angiography (CCTA)

Coronary computed tomography angiography (CCTA) is one of the diagnostic modalities which has been quickly developing over the recent years. To perform it a patient

is laid down on the table of CT scanner, a dye is injected through an intravenous line in patient's hand or arm and several images at different angles are taken by the scanner. Images are later processed to create the final 3D heart scan. CCTA is recommended as an initial test to establish whether patient suffers from CAD according to the guidelines proposed by both European and American cardiology societies [41, 42]. Recent studies have shown that CCTA is a highly accurate non-invasive modality which has been proven to be as effective as its invasive counter-part, conventional coronary angiography (CCA) [43, 44].

4.1 ImageCAS

The ImageCAS [2] dataset consists of 1000 3D CCTA heart images obtained by Siemens 128-slice dual-source scanner which have later been annotated by the experts to indicate coronary artery. Data was collected based on clinical cases from Guangdong Provincial People's Hospital. Included are 414 females and 586 males with the average age of around 58 years. Only people over 18 years old and with documented heart diseases connected to coronary artery were eligible. Each image is of $512 \times 512 \times (206-275)$ voxel size, with planar resolution of $0.29-0.43 \text{ mm}^2$ and $0.25-0.45 \text{ mm}$ spacing. For each scan, there are coronary artery labels assigned by two radiologists. The dataset is one of the first publicly accessible of this size and has been widely used in various research.

Each scan has been additionally propagated through TotalSegmentator [1] to obtain segmentation of the major heart structures. TotalSegmentator is an open source model which allows for segmentation of over 100 anatomic structures from CT images. It does not cover coronary arteries, however, it is helpful in providing an overview of the heart anatomy and exploring relationship between coronary artery and other anatomical structures. An example of heart structure with coronary artery segmentation from ImageCAS, and other segmentations generated by TotalSegmentator, is shown in figure 4.1.

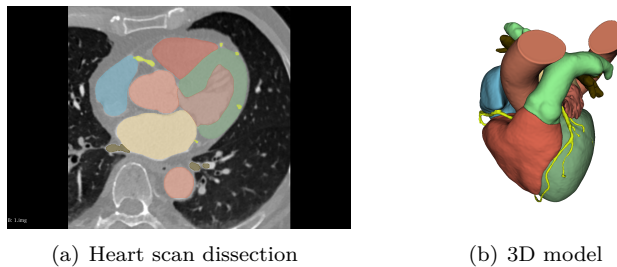


Figure 4.1: Example of an image from ImageCAS dataset. The segmentation generated by TotalSegmentator has been overlaid on the image. Colours indicate different anatomical structures. (a) shows a dissection in one of the dimensions, while (b) shows 3D model of heart anatomical structures with coronary artery marked in yellow.

The dataset suffers from several limitations. Most of images, having various quality issues making the artery segmentation anatomically implausible, were excluded from

the original ImageCAS dataset. That resulted in a total of 762 remaining scans. Still, the remaining images suffered from considerable issues. There are examples of missing small parts of the branches, or even whole parts of the coronary tree, in the segmentations of coronary arteries. Figure 4.2 shows three examples of such issues. In (a), a big part of what seems to be an LCX is missing from the segmentation. In (b), half of the coronary tree: LAD and its subbranches have been completely omitted in the segmentation. In (c), part of the segmentation does not correspond to the actual coronary tree and meanders into other heart structures. Additionally, it can be seen that in some cases it is difficult to determine specific endpoints of the branches, as they get narrower and less visible due to the contrast washing out. All those limitations can impact an algorithm and bias it towards the incorrect data, as well as make it difficult to evaluate its true performance.

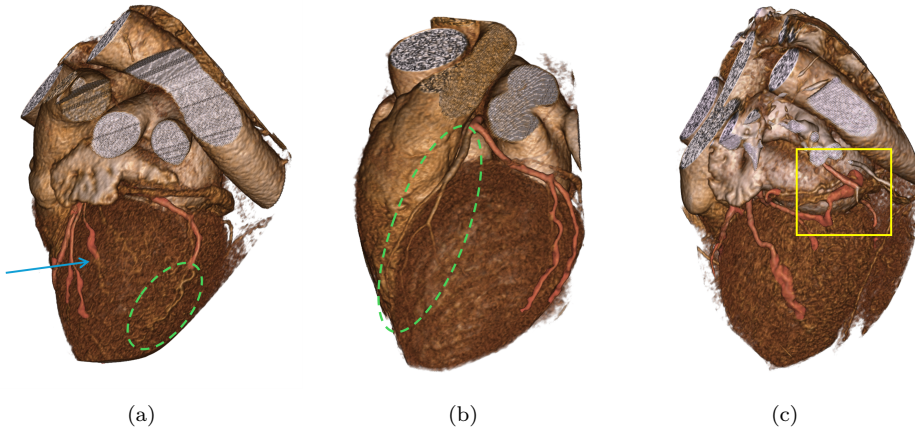


Figure 4.2: Examples of incorrect coronary artery segmentations pictured with heart volume. Segmentations provided in ImageCAS are highlighted in pink. Green circles show parts of the coronary tree that are missing from the ground truth segmentation. Blue arrow points to a branch in which the endpoint might have been incorrectly determined. Yellow rectangle shows part of the segmentation not corresponding to the coronary artery.

Another limitation concerns the quality of the scans themselves. The 128-slice scanner used to obtain the images in ImageCAS, due to its technical constraints, cannot capture the entire heart structure in one rotation. Thus, multiple acquisitions are needed in order to produce an image of the entire heart. That can lead to step artifacts due to cardiac or patient's motion. Step artifacts in ImageCAS can be found throughout all cardiac structures, but due to the nature of this project, they pose the biggest problem when they are found in coronary arteries, as they can be

detrimental to the algorithm's performance. Figures 4.3 and 4.4 show examples of step artifacts in LCA and in other heart structures, respectively.

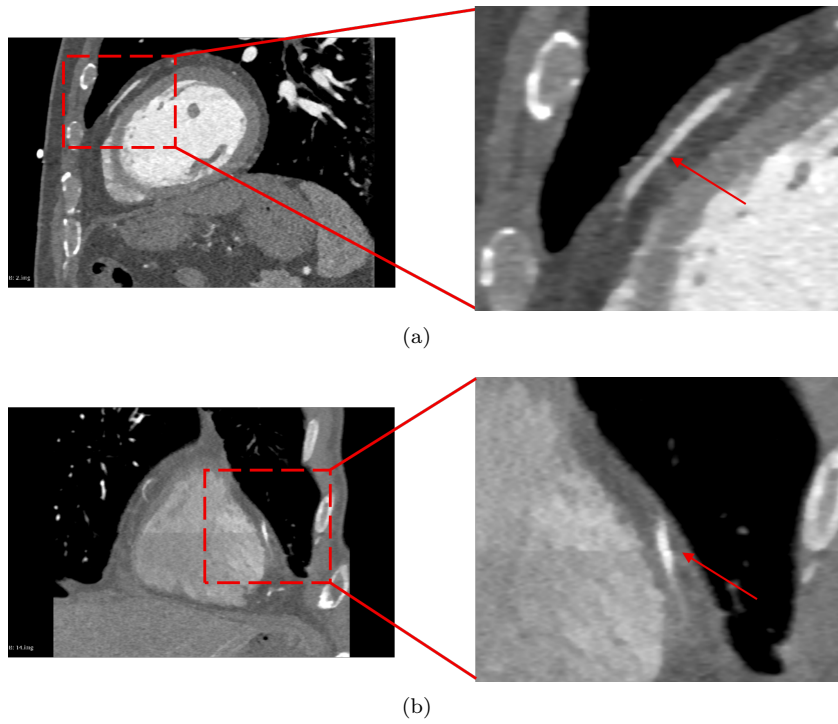


Figure 4.3: Examples of step artifacts in LCA. In (a), a narrow band with different intensity than in the rest of LCA can be observed. In (b), step artifact caused misalignment between two parts of LCA.

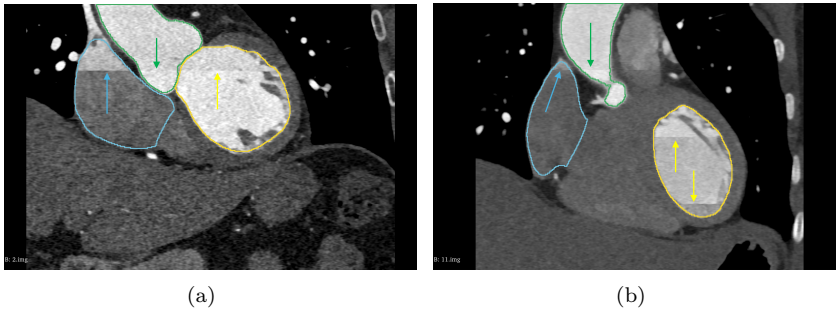


Figure 4.4: Examples of step artifacts in other heart structures than LCA. Blue colour indicates right atrium, green one - aorta, and yellow - left ventricle. In all cases a sudden change of intensity is observed. In left ventricle in (b) it can be noticed twice.

4.2 Copenhagen General Population Study

Apart from ImageCAS dataset, the algorithm’s generalisation abilities were tested on a separate dataset. The dataset is a part of Copenhagen General Population Study (CGPS) [45]. CGPS is a follow-up of the Copenhagen City Heart Study (CCHS) [46], that was set up in 1975 in order to research prevention methods of heart diseases. In the following years, the research was extended by many new aspects, e.g. pulmonary or eye diseases. CGPS has been established in 2003 as an expansion of CCHS, including over 100000 participants aged 20 to 100. Part of the study focusing on CCTA scans was established in 2010 and included over 10000 participants aged 40 or more. The dataset used for this project is a subset of CGPS and comprises 939 scans of asymptomatic individuals. It includes segmentations of left coronary trees automatically extracted with nnU-Net model [47], corrected manually by one expert.

The CPGS dataset was used in order to evaluate the algorithm’s performance and generalisation abilities on new data. As different CT machines produce scans of different qualities, an algorithm’s performance can also differ based on the data used. It is therefore vital to understand if an algorithm trained on one dataset would be suitable for different datasets as well.

5.1 Data preprocessing

In our project we have used a curated version of ImageCAS dataset [2] consisting of 762 images. For the purpose of training models and tuning the tracing algorithm, the images were split between training, validation, and testing datasets consisting of 450, 150 and 162 heart scans respectively.

Each image and the corresponding segmentation has been processed so that it can serve as an input to the relevant models, or to be used in a tracing algorithm. During data processing, each scan has been associated with the following information:

- starting and end points of the coronary artery tree,
- centerlines,
- branch points' locations.

The centerlines are not directly available in the ImageCAS dataset - only the coronary artery meshes are provided there. Hence, the centerlines were extracted given the available segmentations in a following process:

1. Both left and right coronary arteries arise from the aorta above the left and right cusps of the aortic valve, correspondingly [48]. Thus, the starting points were found as an overlap between the segmentations of aorta and the dilation of segmentations of coronary arteries. Center of the overlap was then defined as a starting point.
2. VTK [49] function `vtkDijkstraGraphGeodesicPath` was used to find end points in each coronary artery tree. The function uses Dijkstra's algorithm to find shortest geodesic paths between given points - in our case between the starting point and points on the surface of coronary artery segmentation. End points were defined as surface points that correspond to the local maxima of geodesic distances in small regions.
3. Having coronary artery surfaces, start and end points, `vmtkcenterlines` method from VMTK [50] library was used to extract the centerlines. This method outputs a set of branches, where each branch consists of set of points, that together form a coronary artery tree. Points where different branches merge together serve as branch points. Additionally, each point i is assigned a scalar value corresponding to the radius r_i of a maximum inscribed sphere. The results are shown in figure 5.1(a). Since the algorithm is not precise in areas where arteries are narrow (e.g. in stenoses), the segmentation was artificially dilated in such

regions to ensure proper centerline extraction. This resulted in overestimated spheres (figure 5.1(b)). Nevertheless, the results were accurate enough for the goal of branch spawning.



Figure 5.1: Example of LCA segmentation (in pink) together with maximum inscribed spheres (in yellow). (a) shows the whole coronary tree, whereas (b) shows an overestimation of the spheres due to the narrowing of artery structure.

After centerline extraction, each point is assigned a value corresponding to a distance from a given point to a starting point in millimetres. Thus, the distances are monotonically increasing, reflecting the flow of blood through arteries. An example of such heatmap is presented in figure 5.2.

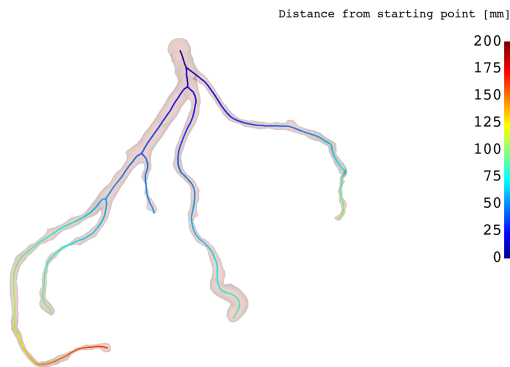


Figure 5.2: Example of LCA centerline with distances heatmap together with coronary artery segmentation.

Additionally, for the branch point detection model use case, we assign a likelihood of each point on the centerline being a branch point. We define an influence radius of 5 millimetres. If a given point lies outside of influence radius from a ground truth branch point, the likelihood assigned to it is equal to zero. Otherwise, we calculate the probability density v_i for a point i having distance to the branch point d_i using a Gaussian distribution, i.e. $d_i \sim \mathcal{N}(0, 2^2)$, and $v_i = f(d_i) = \frac{1}{\sqrt{2\pi \cdot 2^2}} e^{-\frac{(d_i^2)}{2 \cdot 2^2}}$. A heatmap of branch point likelihoods is presented in figure 5.3

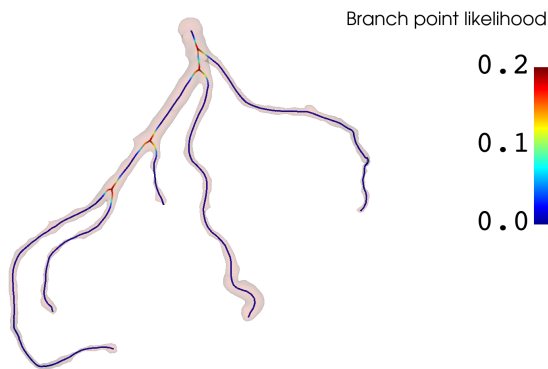


Figure 5.3: Example of LCA centerline with likelihoods of points being a bifurcation.

5.1.1 Image processing

Having the preprocessed data, during the training of the models, as well as in the tracing algorithm, we perform several steps of image processing.

Original image scans are non-isotropic, meaning that they have different spacing in different axes. This can cause problems when training convolutional neural networks, since image patches with the same number of voxels extracted from two different scans may result in differing representations of heart scan volume. This in turn makes it difficult for the model to learn appropriate features from the dataset and generalize well to the new data. To avoid potential issues, all scans are resampled to the isotropic pixel spacing with voxel spacing equal to 0.5 mm.

Additionally, we perform HU augmentation on the images. Iodinated CT contrast corresponds to HU range of 100 – 600 [51]. In order to capture not only the lumen, but also the calcified plaque and surrounding soft tissue, we defined our HU range as -200 to 1200. At the same time, having in mind that the reference values might differ

between individuals, we defined a minimum HU range as $[-800, -200]$ and maximum HU range as $[1200, 2000]$. For each sample, we took a random value from minimum and maximum HU ranges and clipped all other voxel intensities to those values. Then, we normalized the intensities to $[-1, 1]$ scale according to the formula:

$$I_{\text{after}} = (I_{\text{before}} - HU_{\min}) \frac{I_{\max}^{\text{new}} - I_{\min}^{\text{new}}}{HU_{\max} - HU_{\min}} + I_{\min}^{\text{new}} \quad (5.1)$$

where I_{before} , I_{after} are intensities before and after normalization, $HU_{\min}$ and $HU_{\max}$ are HU values sampled from minimum and maximum HU ranges, $I_{\min}^{\text{new}}$ and $I_{\max}^{\text{new}}$ are new minimum and maximum intensity values equal to -1 and 1 , correspondingly.

5.2 Centerline tracer

One of the key components of the centerline tracing algorithm is being able to trace a single centerline. The approach used in this project is a modification of a solution used in Radutoiu's thesis [13]. We incorporate a so called tracer, that after being placed in a specific voxel, can move to one of $N = 26$ adjacent voxels. In other words, we define a set of possible directions $\mathcal{D}$, in which the tracer can follow, as:

$$\mathcal{D} = \{\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_{26}\}, \quad \text{where } \mathbf{d}_i \in \{-1, 0, 1\}^3, \mathbf{d}_i \neq \mathbf{0} \quad (5.2)$$

Radutoiu proposed a convolutional neural network (CNN) that outputs a likelihood of each direction being the best one - a direction with the highest value was chosen as the next location of the tracer. Our modified approach incorporates a similar model, but with the output being a direction quality score that a step in each direction would achieve, according to the function defined below. This modification, which essentially changes the CNN from a classification to regression type, provides more flexibility. Having an approximation of quality score for each direction allowed us to experiment with incorporating it into branch spawning when evaluating potential path candidates, as described in section 5.6.1.

5.2.1 Direction quality score function

For the centerline tracing we used a direction quality score function that evaluates how good is a direction of a step taken from current point to the next one, inspired by reward function in Ekner's thesis [52]. Its purpose is to encourage taking actions which keep the tracer close to the centerline, as well as to follow it in the direction of blood flow. For that purpose, it incorporates distances of the current position to the centerline D_{to} and along the centerline D_{along} :

$$D_{\text{to}}(\mathbf{p}) = \|\mathbf{p} - \mathbf{g}_{\mathbf{k}}\|_2 \quad (5.3)$$

$$D_{\text{along}}(\mathbf{p}) = v_k = \frac{\|\mathbf{g}_k - \mathbf{g}_{k-1}\|_2}{d_{\text{total}}} \cdot v_{k+1} + \frac{\|\mathbf{g}_{k+1} - \mathbf{g}_k\|_2}{d_{\text{total}}} \cdot v_{k-1}, \quad d_{\text{total}} = \|\mathbf{g}_{k+1} - \mathbf{g}_{k-1}\|_2 \quad (5.4)$$

where:

- $\mathbf{g}_k$ - the closest point on the true centerline to current location $\mathbf{p}$,
- v_k - cumulative distance along the centerline from the root of the artery.

D_{along} uses an interpolation to rectify the possibility of the closest point on the centerline laying between the two previously defined points on the centerline.

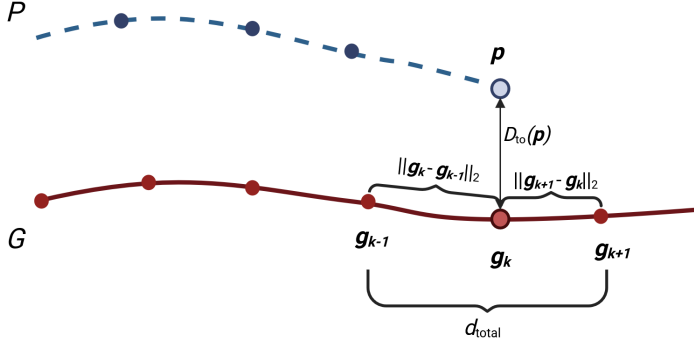


Figure 5.4: Visualization of components used for direction quality score function calculation. G and P correspond to ground truth and predicted centerline, respectively. Created with [53].

The proposed direction quality score function, assessing the quality of each move, incorporates D_{to} and D_{along} , and is defined as:

$$R = \begin{cases} \omega \alpha (D_{\text{to}}(\mathbf{p}) - D_{\text{to}}(\mathbf{p}_{\text{next}})) & \text{if } D_{\text{to}}(\mathbf{p}_{\text{next}}) > \tau \\ \omega (\alpha (D_{\text{to}}(\mathbf{p}) - D_{\text{to}}(\mathbf{p}_{\text{next}})) + (D_{\text{along}}(\mathbf{p}_{\text{next}}) - D_{\text{along}}(\mathbf{p}))) & \text{otherwise} \end{cases} \quad (5.5)$$

As an agent moves from its current location, $\mathbf{p}$, to the next one, $\mathbf{p}_{\text{next}}$ the distances to the centerline and along the centerline are calculated for both points to evaluate the score function value. If a distance to the centerline in the new state exceeded

threshold denoted by $\tau = 1$ mm, only the difference in the distances to the centerline between two points is considered. If, however, the distance to the centerline in the new state is within the limit, the direction quality function takes into account change in distance along the centerline as well. Parameter $\alpha = 2$ is used to emphasize the proximity to the centerline rather than distance along it. Parameter $w = \frac{1}{\|\mathbf{d}\|}$, where $\mathbf{d}$ - direction between $\mathbf{p}$ and $\mathbf{p}_{\text{next}}$, is used to account for bias towards selecting diagonal directions caused by them having larger step lengths. In other words, choosing a diagonal direction results in a greater distance between $\mathbf{p}$ and $\mathbf{p}_{\text{next}}$, and without factor w it would lead to a higher direction score.

Direction that corresponds to the largest score is considered the best one.

5.2.2 Model architecture and training

In order to evaluate the direction that the tracer should follow when being in a specified location, we incorporated a CNN that estimates a direction quality score obtained by following each of the possible directions. The input to the network is an image patch of dimensions $41 \times 41 \times 41$ voxels, centred at this specified location. Image patch dimensions correspond to $20.5 \text{ mm} \times 20.5 \text{ mm} \times 20.5 \text{ mm}$ in physical dimension, ensuring that a patch covers the entire width of a coronary artery as well as its immediate surroundings.

The convolutional neural network consists of 4 convolutional 3D blocks, that serve as a feature extraction part, inspired by [54], and a regression part. Each block comprises two 3D convolutional layers with kernel size equal to $(3, 3, 3)$ and padding equal to 1. Both are followed by batch normalization and ReLU activation function. Block is wrapped with max pooling. After the convolutional 3D blocks we apply adaptive average 3D pooling. Next, in the regression part, we use 3 fully connected layers - first two followed by a ReLU activation function. The final output vector has length of 26. Each element of the vector denotes the predicted direction score obtained when following a given direction. Schema of the model architecture can be found in figure 5.5.

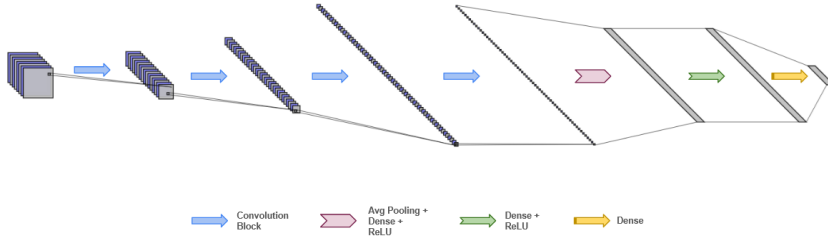


Figure 5.5: Simplified schema of model architecture. Architecture is represented in 2D for simplicity. Network takes a collection of $41 \times 41 \times 41$ patches and outputs a vector with direction quality scores for 26 possible directions. 3D patches, used in convolutions, are represented by square shapes. Patch size is reflected by the square size. Three final outputs are one dimensional vectors. Colour coded arrows represent different types of blocks or layers.

During the training, points are randomly translated off the centerline by up to 2.5 mm, as described by Radutoiu [13], with probability of 80% that the point is going to be translated. Due to the computational limitations, in each epoch we sample 55 points per scan, filling the memory. Having 450 scans in the training set, a memory capacity is thus equal to 25000. Before a new epoch begins, 10 runs through memory are performed. This procedure ensures that that the training process is sped up to adhere to computational constraints, while creating as varied set as possible to avoid overfitting. The training parameters are described in table 5.1.

Training setting	Value
loss function	Mean Squared Error Loss
optimizer	Adam
number of epochs	50
runs through memory	10
batch size	64
learning rate	10^{-4}
weight decay	10^{-3}

Table 5.1: Overview of the training setup for centerline tracer.

5.2.3 Reverse tracing

The algorithm used in this project for tracing entire coronary tree consists of three key steps:

1. Tracing the first centerline starting from the overlap point with aorta,
2. Detecting branch points on the tracked centerline,
3. Trace new centerlines from the vicinity of detected branch point.

Steps 2 and 3 are repeated until no new branch points are detected. More details on that in section 5.7. The proposed solution has one major drawback: parts of the coronary tree between the detected branch points and starting points of new centerlines are not tracked, resulting in gaps in the traced coronary tree. To this end, we explored a possibility of incorporating a reverse tracer, that starts tracing from the last point of a predicted centerline and proceeds with tracing in the opposite direction to the blood flow. For this purpose, a direction quality score function was modified in a following way:

$$R = \begin{cases} \omega\alpha(D_{\text{to}}(\mathbf{p}) - D_{\text{to}}(\mathbf{p}_{\text{next}})) & \text{if } D_{\text{to}}(\mathbf{p}_{\text{next}}) > \tau \\ \omega(\alpha(D_{\text{to}}(\mathbf{p}) - D_{\text{to}}(\mathbf{p}_{\text{next}})) + (D_{\text{along}}(\mathbf{p}) - D_{\text{along}}(\mathbf{p}_{\text{next}}))) & \text{otherwise} \end{cases} \quad (5.6)$$

This slight modification takes place in case the distance of a new state ($\mathbf{p}_{\text{next}}$) to the centerline does not exceed the threshold τ , and ensures that the tracer is now guided to take actions along the centerline, but in the direction opposite to the blood flow. Incorporating the reverse tracing into the final algorithm could eliminate an issue of gaps in the traced centerlines.

Reverse tracer network has the same model architecture and training strategy as the regular one.

5.3 Branch point detector

Tracer alone is designed to follow one centerline until the terminating conditions are met. Because of that, we introduce a branch point detector which acts as a tool for indicating where the already traced centerline has bifurcation points so that the tracer can follow subsequent branches.

The branch detector model is a slightly modified version of the one proposed in Radutoiu's thesis [13]. The model architecture follows the one described in section 5.2.2 with the modified output layer, as in this case the model outputs only one value, the likelihood of a point being a branch point. The ground truth probabilities are obtained as described in section 5.1. A lot of points lay far from any bifurcation point,

creating a heavy class imbalance. To remedy that, a sampling weight is introduced favouring samples close to bifurcation points.

The trained model allows to create a heatmap of bifurcation point likelihoods given a coronary artery centerline. In the neighbourhood of true branch points the likelihoods can be high for several points on a predicted centerline. To indicate where within such area the branch point actually is, we implemented a solution proposed by Radutoiu [13]. The method first checks a range of heatmap values on a predicted centerline to establish if any branch points are present on it. If so, it finds the point with the highest value, and calculates the weighted average location within its neighbourhood. Training is performed with a similar procedure as described in section 5.2.2.

5.4 Artery radius estimator

An important part of the algorithm is finding new tracing directions from detected bifurcation points. Having insight into the size of an artery in a branch point helps with deciding how much space to consider when looking for branch candidates. This consideration led to the development of an artery radius estimator. The process of finding new directions from branch points incorporating radius estimations is described more in detail in section 5.6.

The goal of an artery radius estimator is to approximate a radius, r_{est} , for a given point on a centerline, $\mathbf{p}$. The form of input data, as well as the architecture of the CNN are the same as described in section 5.2.2. The only difference lies in the output layer: here we use a layer with a single output denoting r_{est} . The training follows the same procedure as described in section 5.2.2, with one slight modification. Since the data complexity in the radius estimation task is significantly lower than in centerline tracing, and the task itself is therefore also simpler, we decided to perform only a single run through the memory.

5.5 Artery likelihood estimator

Another important part of the algorithm is the artery likelihood estimator. The convolutional neural network model is aimed at predicting the likelihood of a tracer being inside a coronary artery. It is utilized both in terminating conditions and branch spawning.

The model architecture follows the one described in section 5.2.2, however, the output layer has been modified. The last fully connected layer outputs just one value and, since it is the likelihood that we aim at predicting, it is passed through a sigmoid activation function to squash the results to the unit interval. The training procedure is similar as in section 5.2.2. The training examples are also translated away from the centerline, similarly as for the centerline tracer, but up to 4 mm. This produces

roughly equal distribution of points in and outside the coronary artery. For the loss function a binary cross entropy loss has been chosen. The final output of the model is evaluated against a binary ground truth label indicating whether a given state lays within the coronary artery. The binary ground truth labels have been obtained by comparing centre of the input image patch against the segmentation. The simplified process has been visualized in figure 5.6.

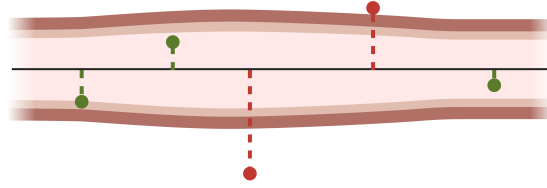


Figure 5.6: Simplified process of evaluating ground truth labels of points sampled around the coronary artery for artery likelihood estimator. Colours of the points indicate ground truth labels with red meaning outside, and green inside the artery segmentation. Created with [53].

5.6 Branch spawning

Branch detector allows for estimating locations of bifurcation points. Given the branch point, the objective of branch spawning solution is to determine which direction should the tracer move towards to follow a yet untraced branch. This section presents several approaches which we investigated.

5.6.1 Direction-score-based exploration

Direction-score-based exploration is a method of determining directions of branches based on the accumulated direction score values at certain paths. The procedure comprises three key steps:

1. Establishing a set of directions to explore,
2. Tracing in each direction for a number of steps and calculating accumulated direction quality score,
3. Filtering out paths based on a number of conditions.

Establishing exploring directions

A popular method of distributing points on a sphere is a latitude-longitude lattice [55], where points are located on the intersections of latitude and longitude lines. A drawback of this method is that points are distributed in a highly anisotropic manner and are clumped together near the poles. To avoid those issues, we used a Fibonacci sphere algorithm [55]. In this method, points are generated on a unit sphere based on the golden ratio value, helping to achieve isotropic spacing. A visualization of both methods has been shown in figure 5.7. For the goal of branch spawning, we generated 50 points on a Fibonacci sphere, each corresponding to a potential branch direction $\mathbf{d}_1, \dots, \mathbf{d}_{50}$.

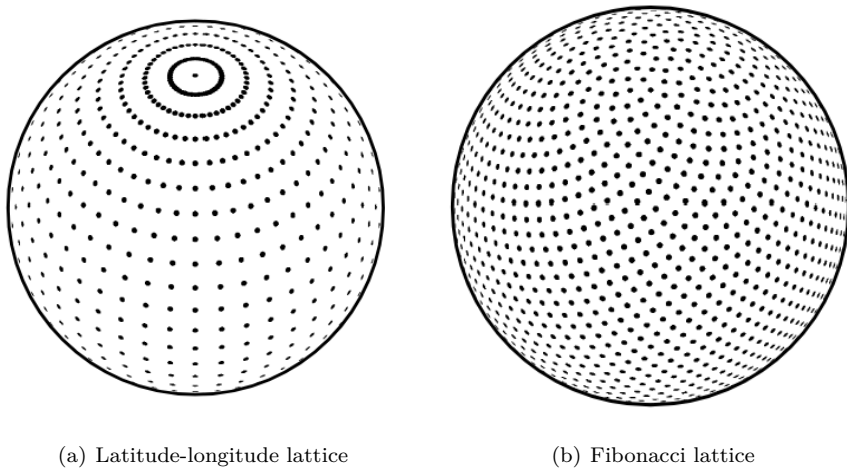


Figure 5.7: Comparison of points distribution on a sphere. Points of Fibonacci sphere are distributed fairly evenly comparing to the latitude-longitude one. Modified from [55].

Tracing in potential directions

After establishing potential directions, we estimate a radius value r_{est} at the bifurcation location, and start tracing in a distance of $1.5 \cdot r_{\text{est}}$ from it in each potential direction. Starting tracing too close to the branch point could result in the tracer following already traced centerline and missing potential branches. Paths in each direction are traced for 20 steps or until the likelihood of being inside an artery is too low (described in detail in section 5.7.1). For each path candidate, accumulated direction score estimated by the centerline tracer is calculated.

Filtering path candidates

Out of all traced candidate paths, only those having Euclidean length greater than 2 mm are considered.

Denoting $\mathbf{p}_s$ and $\mathbf{p}_e$ as starting and ending points of a candidate path, respectively, we define a direction of a path as:

$$\mathbf{d}_{\text{path}} = \frac{\mathbf{p}_e - \mathbf{p}_s}{d(\mathbf{p}_e, \mathbf{p}_s)} \quad (5.7)$$

where $d(\mathbf{p}_e, \mathbf{p}_s)$ is a Euclidean distance between $\mathbf{p}_e$ and $\mathbf{p}_s$, serving as a normalizing factor. If, for a given explored direction $\mathbf{d}_i$, $\mathbf{d}_{\text{path}} \cdot \mathbf{d}_i \leq 0.2$, we assume that the tracer wandered too much away from the original direction, and the path candidate is discarded.

All paths for which $\mathbf{p}_e$ is closer than 2 mm to any point on already traced centerlines, are also discarded, as they are too near and therefore too similar to already traced centerlines.

Finally, path candidates, for which distances between the ending points are smaller than 5 mm, are compared to each other. For each set of such paths the direction with highest average direction score is chosen as the final candidate, with the rest of them being discarded.

After performing the filtering procedure, a number of found directions $\mathbf{d}_i$ shifted by $1.5 \cdot r_{\text{est}}$ serve as starting points of new branches.

5.6.2 Likelihood-sphere-based spawning

Alternative approach considered was centred around the idea of utilizing artery likelihood estimator to establish which direction is the most likely to be following an artery. There are two steps included in this approach: building a direction likelihood map and, determining the directions in which the agent should continue tracing based on it.

Artery likelihood sphere

Once a bifurcation point $\mathbf{b}$ is established, r_{est} is obtained from the radius estimator. Then, we establish 400 rays corresponding to 400 directions $\mathbf{d}_1, \dots, \mathbf{d}_{400}$ on the Fibonacci sphere [55]. Along each ray we evaluate artery likelihood on 11 evenly spaced points defined as:

$$\mathbf{p}_{ki} = \mathbf{b} + \mathbf{d}_i \cdot r_{\text{est}} \cdot k, \quad \text{where } k \in K = \{1, 1.25, \dots, 3.5\} \quad (5.8)$$

Here, $\mathbf{p}_{ki}$ is a point corresponding to ray $\mathbf{d}_i$ shifted from $\mathbf{b}$ by $r_{\text{est}} \cdot k$. The closest vicinity of the bifurcation point is not considered. Finally, the accumulated likelihood value along each ray is calculated according to the following equation:

$$a_i = \sum_{k \in K} f(\mathbf{p}_{ki}) \quad (5.9)$$

where $f(\mathbf{p}_{ki})$ is an artery likelihood in point $\mathbf{p}_{ki}$.

This results in a discrete spherical likelihood map of 400 points laying on the surface of the sphere with radius equal to $3.5 \cdot r_{\text{est}}$. Therefore, the value a_i of each point reflects the certainty of following the branch in a particular direction. The final likelihood values have been normalized to a unit interval. A simplified visualization of this process is shown in figure 5.8.

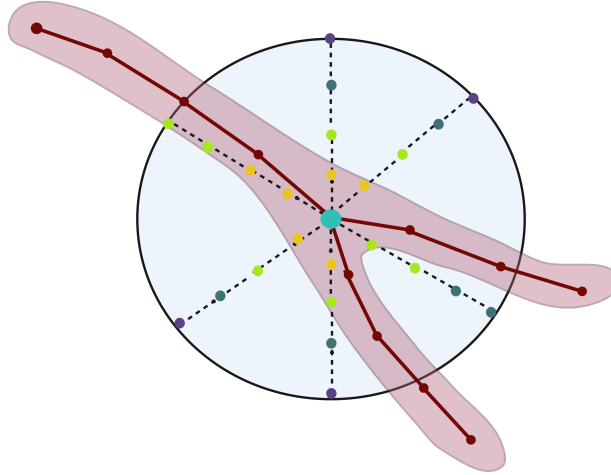


Figure 5.8: Process of a building a directional likelihood sphere around the branch point. Visualization is reduced to 2D for aesthetic appeal. The colours of points along the rays coming out of branch point indicate the likelihood of point laying inside an artery. The darker the shade, the less likely the point is to be inside an artery. Values along the rays are later summed and normalized. Created with [53].

von Mises Fischer mixture model

Having created the directional likelihood map, the next part is to extract directional vectors pointing towards the highest artery likelihoods. This task can be approached in various ways. Most likely the simplest solution is to use k-Nearest Neighbours algorithm, however, this would restrict the optimal direction to be identical to one of the sampled rays and, given that the approach considers only 400 rays, it is very

likely that the true optimal direction would lay somewhere in between. Therefore, it has been decided to employ a model-based approach.

The likelihood space lays on a surface of a three dimensional sphere which makes von Mises-Fisher (vMF) distribution [56] a perfect candidate for the model. Von Mises-Fisher distribution is a directional distribution defined on a unit hypersphere. It is often perceived as an equivalent of a normal distribution on a sphere. It is characterized by the following functional form:

$$f_p(\mathbf{x}; \mu, \kappa) = C_p(\kappa) \exp(\kappa \mu^\top \mathbf{x}) \quad (5.10)$$

where p denotes dimensions of the unit vectors, κ is the concentration parameter, μ represents mean direction, while $C_p(\kappa)$ is the normalization constant of the following form:

$$C_p(\kappa) = \frac{\kappa^{p/2-1}}{(2\pi)^{p/2} I_{p/2-1}(\kappa)} \quad (5.11)$$

which in a 3-dimensional case simplifies to:

$$C_3(\kappa) = \frac{\kappa}{4\pi \sinh \kappa} = \frac{\kappa}{2\pi(e^\kappa - e^{-\kappa})} \quad (5.12)$$

Importantly the support of the distribution is the hypersphere, more specifically 2-sphere, defined as:

$$S^2 = \{\mathbf{x} \in R^2 : \|\mathbf{x}\| = 1\} \quad (5.13)$$

which is precisely what the likelihood map represents.

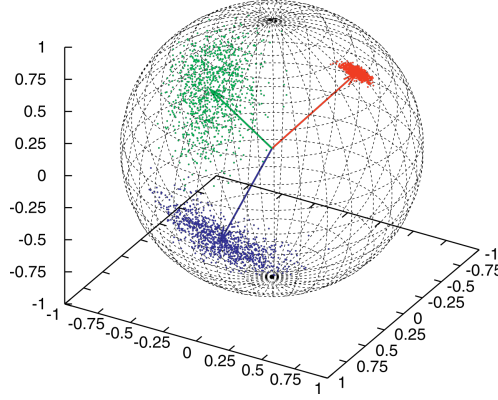


Figure 5.9: The image shows three colour coded distributions fitted to point clouds laying on the sphere. The particular distribution used in this case is Kent distribution of which a special case is von Mises-Fisher distribution. The purpose of the image is to illustrate the models fitted on spherical data. Image from [57].

Having chosen an appropriate distribution, a mixture model of several components, each following the von Mises-Fisher distribution, is created. We chose a number of components to lay between 2 and 4, where 2 means that an analysed branch point is a false positive and is located in the middle of an artery, whereas 4 means that it corresponds to trifurcation. In very rare cases quadrifurcations are observed in a population [58]. Since they are extremely unusual, this possibility was omitted for the purposes of this project. The constructed mixture model is therefore finite and can be represented as follows:

$$f(\mathbf{x}) = \sum_{j=1}^J \pi_j \cdot f_3(\mathbf{x}; \mu_j, \kappa_j) \quad (5.14)$$

where π_j denotes the likelihood of a point belonging to j -th component.

Lastly, an optimization is performed by minimizing least-squares loss function and therefore the target can be defined as:

$$L = \sum_i^{400} \left(a_i - \sum_{j=1}^J \pi_j f_3(\mathbf{x}; \mu_j, \kappa_j) \right)^2 \quad (5.15)$$

The function is optimized numerically using a standard L-BFGS method for unconstrained optimization problems [59].

Once the directions are established, the tracer starts following each of them from the spawn point offset by $1.5 \cdot r_{\text{est}}$ from a branch point, and then continues tracing in a regular manner. The procedure of filtering candidate paths is much simplified compared to the one applied for direction-score-based spawning described in section 5.6.1. The only condition that is checked is whether the end point $\mathbf{p}_e$ of the path candidate is at least 2 mm away from any already traced centerline.

Model selection criteria

Having a framework for fitting our model for a given number of components, it is important to decide how many components should be fitted on each sphere. This will of course vary between the spheres so each sphere should be considered separately. To choose an optimal number of directions, first a mixture model is fitted for each number of components and then all models are evaluated to establish which one reflects the data best. Relying on the loss values would be sub-optimal since the loss can often simply decrease with the number of components increasing, as they make the model more flexible. This is where Akaike and Bayesian information criteria [60, 61] can be of use as they penalize the model for too high complexity (number of free parameters). Criteria can be calculated in the following way:

$$\text{AIC} = 2p - 2\ln(\hat{L}) \quad (5.16)$$

$$\text{BIC} = p\ln(N) - 2\ln(\hat{L}) \quad (5.17)$$

where p - number of free parameters, N - number of observations, $\hat{L}$ - maximized likelihood for a given model. Note that especially for models with large number of samples, the BIC model will penalize higher complexity more severely than AIC.

Initialization methods

Any kind of numerical optimization relies on initialization of the starting points. Poor initialization can influence the final solution so the correct initialization technique should be selected carefully. In the optimized mixture model mixture weights, concentration parameters, and mean directions need to be initialized. Since mean directions are the most important for our use case, we focus on improving only their initialization and randomly initialize the remaining parameters. The simplest solution is to randomly, independently draw number of points on the sphere, however, this approach is often not reliable, as the final results may vary depending on the run. We considered the following two approaches.

Minimization of electrostatic potential energy The idea behind this approach originates from the paper published in 1904 by J.J. Thomson [62] in which the author states the problem of arranging electrons on a sphere in such a way that the

total electrostatic potential energy is minimized. The energy is minimized when the electrons are as far from each other as possible. In other words, the solution to the problem is to find a way of spreading the objects evenly on a sphere. This can be well utilized to initialize points in our approach since in majority of cases the branches are relatively far from each other. To achieve that, the following energy function needs to be minimized:

$$E = \sum_{i < j} \frac{1}{\|\mathbf{x}_i - \mathbf{x}_j\|} \quad (5.18)$$

Note that we introduce another optimization problem and this in turn again requires initialization. Nevertheless, this solution ensures that the points are evenly spread even if their placement is still random. To rectify this further, we additionally anchor one of the points on the sphere by selecting a global likelihood maximum which should lay very close to one of the branch directions. The only optimized points are therefore the remaining ones. The function is again optimized using the same framework as in case of loss function [59].

k-Nearest Neighbors We previously advocated for not following the kNN inspired solution for finding the maxima on the sphere, however, it still can be of use to resolve the initialization problem. Our idea is to use kNN to find local maxima and consider them starting points for fitting the von Mises-Fisher model. The maxima can be found by scanning through the neighbourhood of each point on the sphere and checking if the likelihood value of the considered point is higher than the likelihood value of any of its neighbours. The algorithm is quick, requires only one pass and is deterministic which eliminates the randomness factor. More importantly, this method eliminates the need for comparing mixtures with different numbers of components as the number of local maxima found equals number of fitted directions. The solution still requires careful choice of number of neighbours considered and can result in finding too many local maxima. Thus, in addition, we propose a relative threshold condition which accepts the local maxima only if they are not lower than the global maximum times the τ factor. Mathematically, the problem can be stated in the following way:

$$\text{Local maxima: } a_i > a_j \quad \forall j \in \mathcal{N}_k(i) \quad \text{and} \quad a_i > \tau \cdot \max_m a_m \quad (5.19)$$

where $\mathcal{N}_k(i)$ represents the neighbourhood of k points around point i .

5.7 Coronary tree tracing algorithm

This section presents an overview of an algorithm for tracing the whole left coronary tree. The entire algorithm is based on an idea of alternating tracing single centerlines and detecting branch points on them.

Tracing single centerline is based on predicting the next location of the tracer $\mathbf{p}_{\text{next}}$ given its current location $\mathbf{p}$. This procedure is repeated for a maximum of 1000 steps, unless one of the terminating condition is met earlier (described below). At each new location of the tracer, we check if the terminating conditions are met. If not, and the tracer appears in $\mathbf{p}_{\text{next}}$ for the first time, current location of the tracer is updated as $\mathbf{p} = \mathbf{p}_{\text{next}}$ and tracing continues. If $\mathbf{p}_{\text{next}}$ has already been visited before, momentum procedure is applied (described below) to guide the tracer out of the problematic area, before the tracing is continued. Tracing of a single centerline is presented in figure 5.10

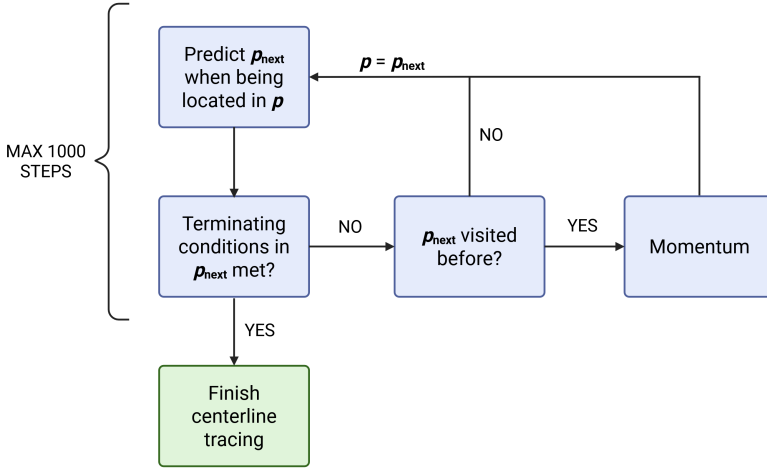


Figure 5.10: Schema of tracing single centerline The procedure is repeated until the maximum limit of steps is reached, or until terminating conditions are met. Created with [53].

Starting points of each centerline are saved in a queue $\mathbf{P}$. At the very beginning of an algorithm, $\mathbf{P}$ comprises only one point, $\mathbf{p}_0$, being a point of overlap between the aorta and coronary artery tree, extracted during data processing. Tracing of each centerline $\mathbf{C}$ starts from a corresponding starting point and it is tracked until one of the terminating conditions is met (described below). Then, each point in $\mathbf{C}$ is passed through a branch detector to identify any possible branch points $\mathbf{B}$. Each of the detected branch points is then used for a branch spawning procedure in order to find the beginning of new branches in a coronary tree, that are then added to the set of starting points $\mathbf{P}$. The procedure is repeated until there are no points in $\mathbf{B}$ and $\mathbf{P}$

left. Schema of the tracing algorithm is showed in figure 5.11.

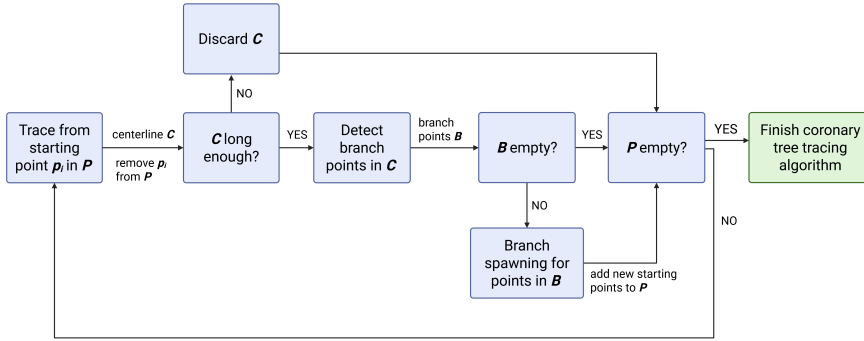


Figure 5.11: Schema of coronary tree tracing algorithm. The algorithm runs until both queue of starting points P and of branch points B are empty. Created with [53].

5.7.1 Terminating criteria

In order to decide when to finish the tracing, we developed a few terminating criteria. An important thing for consideration was that terminating the tracing too late is not as harmful as terminating too early. Terminating too early would result not only in a missing part of a centerline, but also in inability to detect potential bifurcation points on it and subsequent parts of the coronary tree. On the other hand, tracing too far could lead to inaccuracies and the tracer getting lost in areas where coronary arteries are not present any more. Nevertheless, the terminating conditions should lay in balance between capturing as much of coronary artery tree as possible and avoiding unrealistic results.

A following set of terminating criteria was incorporated in the tracing algorithm:

1. Reaching a maximum number of steps, set as 1000,
2. Tracer going out of the image borders after taking a step in a predicted direction,
3. Having an average HU value of voxels in last 5 steps lower than 1,
4. Visiting the same point twice after momentum (described in section 5.7.2),
5. Having too low average likelihood of being inside an artery calculated over the last 3 steps.

Conditions 1 and 2 are straightforward. Criterion 1 guarantees tracing termination if none of other conditions is met. A chosen number of steps is presumed to be enough to trace the entire single centerline. In practice, this condition is rarely enforced. Criterion 2 ensures that the tracing stays within the image boundaries.

Condition 3 ensures that the HU values along the tracked centerlines are high enough. Coronary arteries in CCTA scans, due to the contrast enhancement, reach values of a few hundreds [51], thus voxels with HU values below 1 can be assumed not to lay in the coronary arteries volume.

Condition 4 makes sure that the tracing terminates if the tracer oscillates. If the momentum procedure is applied, then visiting the same point twice means applying momentum to push through a problematic area. If, even with momentum applied, tracer again comes back to the same location, tracing is terminated. In case the momentum procedure is not used, tracing terminates immediately when a given point is visited twice.

Condition 5 is described in more details below.

Too low likelihood of being inside an artery

We determine an artery likelihood threshold for which we terminate tracing as $a_T = 0.25$. Denoting current state of the tracer as $\mathbf{p}_n$, and the output of the artery likelihood estimator for that state as $f(\mathbf{p}_n)$, we define terminating conditions as:

$$\frac{f(\mathbf{p}_n) + f(\mathbf{p}_{n-1}) + f(\mathbf{p}_{n-2})}{3} < a_T \quad (5.20)$$

If the condition from eq. 5.20 is met, the tracing is terminated. The value of the chosen threshold serves as a balance between tracking for too long and missing parts of the coronary tree due to stopping too early. Calculating an average of last three visited points ensures that we do not penalize the tracer if it occasionally goes away from the centerline, and that we need to see a consistent pattern of tracing away from the artery to terminate.

For the initial tracked centerline in a coronary tree we allow the tracer to work for 10 steps before checking the terminating conditions. The reason for that is the segmentation of the coronary tree in regions close to the aorta is of mixed quality, which is why the outputs of the artery likelihood estimator tend to be low in those areas, causing a risk of early stopping and not capturing the entire coronary structure.

For all subsequent centerlines we start checking the terminating conditions after 5 iterations of tracing, leaving the tracer a possibility to track through potentially uncertain regions around bifurcation points.

The algorithm for terminating the tracing based on the likelihood of being in an artery is presented in figure 5.12.

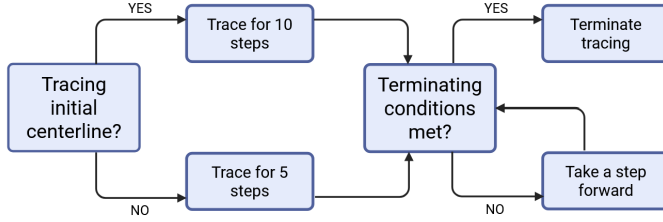


Figure 5.12: Schema of terminating conditions for a single centerline. Terminating criteria are first checked after 10 steps for the initial centerline, and after 5 steps for all subsequent centerlines. Created with [53].

5.7.2 Momentum

Sometimes, when the tracker gets too uncertain in some areas, it comes back to the point visited before, causing an infinite loop. This kind of issue often occurs close to the end of the centerlines. In such cases, simply terminating the tracing would not be too harmful for the algorithm's performance. At the same time, the problem might arise far from the ending points, while the tracker is still localized inside an artery. This could happen due to accumulated plaque or data quality issues. In order to resolve the problem, we implemented a momentum solution, that helps the tracker push through such uncertain areas and continue tracing.

When we encounter the same point twice, we first make sure that the current state is not below artery likelihood threshold, using equation 5.20 with threshold $a_T = 0.25$. Here, $\mathbf{p}_n$ is the first instance of a point visited twice. It means that we do not include points that the tracer was looping over in the calculation of an averaged likelihood.

Next, we calculate an average direction that the tracer was following in the last 5 steps before entering the loop. We consider the first instance of the point visited twice as the last point before entering the loop, and denote it $\mathbf{p}_e$. A point visited 5 steps earlier is denoted $\mathbf{p}_s$. The direction is then calculated as:

$$\mathbf{d}_{\text{avg}} = \frac{\mathbf{p}_e - \mathbf{p}_s}{d(\mathbf{p}_e, \mathbf{p}_s)} \quad (5.21)$$

where $d(\mathbf{p}_e, \mathbf{p}_s)$ is a Euclidean distance between $\mathbf{p}_e$ and $\mathbf{p}_s$ and serves as a normalizing factor.

The new point, from which the tracing algorithm continues, is calculated as:

$$\mathbf{p}_{\text{new}} = \mathbf{p}_e + \sigma_m \cdot \mathbf{d}_{\text{avg}} \quad (5.22)$$

In this project, $\sigma_m = 3$. Its value was chosen to send the tracer far enough so that it does not come back to the same point again, while minimizing the risk of sending

it to other centerlines in the coronary tree our to the regions where coronary arteries are not present.

5.7.3 Reverse tracing

As already mentioned in section 5.2.3, the proposed algorithm suffers from small gaps between the branch points and starting points of new branches. One way to overcome this obstacle would be to manually stitch the missing parts of the tree. In order to find a more robust way, we implemented the reverse tracer that is being set up after the main tracing algorithm finishes. Having the whole coronary tree extracted, tracing starts again, this time from the ending points of each path found before, in the direction of the root. Reverse tracing is performed with the same terminating conditions as the regular one, with one exception - momentum is not implemented in this case, and tracing terminates as soon as the tracer visits the same point twice.

5.8 Tracing evaluation methods

5.8.1 Weighted accuracy

Weighted accuracy is a measure of evaluating a prediction given by a tracer network, being an extension of a regular accuracy metric. Out of 26 possible directions which the tracer can follow, only one, yielding the highest direction quality score, is chosen as a correct one. Nevertheless, all other directions are not equally inadequate. For example, if the true correct direction in which the tracer should go is straight forward, then predicting a diagonal step instead is less detrimental than going in a perpendicular or opposite direction. To account for different quality of predicted directions, weighted accuracy has been introduced. Defining $\mathbf{d}_{\text{gt}}$ as a ground truth direction in which the tracer should take a step from a current location $\mathbf{p}$, and $\mathbf{d}_{\text{pred}}$ as a predicted one, the weighted accuracy can be specified as:

$$w = \max \left(0, \frac{\mathbf{d}_{\text{gt}} \cdot \mathbf{d}_{\text{pred}}}{\|\mathbf{d}_{\text{gt}}\|_2 \|\mathbf{d}_{\text{pred}}\|_2} \right) \quad (5.23)$$

Weighted accuracy is a measure based on cosine similarity. For a given location of the tracer, it is equal to 1 if $\mathbf{d}_{\text{gt}}$ and $\mathbf{d}_{\text{pred}}$ are equal to each other, and 0 if the angle between the two vectors is greater than or equal to 90 degrees. Otherwise, the closer $\mathbf{d}_{\text{gt}}$ and $\mathbf{d}_{\text{pred}}$ are to each other, the closer to 1 the weighted accuracy is.

5.8.2 Evaluation metrics

The main method of tracing evaluation is an **overlap (OV)**, calculated as defined by Schaap et al. [63] All points on the ground truth and on the predicted centerline are

marked as true positive, false positive, or false negative, according to the following criteria:

- a point on the ground truth centerline is considered **TPR** (true positive reference) if its distance to the predicted centerline is lower than the artery's radius at a given point, otherwise it is considered to be **FN** (false negative),
- a point on the predicted centerline is marked as a **TPM** (true positive method) if it's distance to at least one point on a ground truth centerline is smaller than radius at a corresponding reference point, otherwise it is defined as **FP** (false positive)

An illustration of centerline points' classification is shown in figure 5.13.

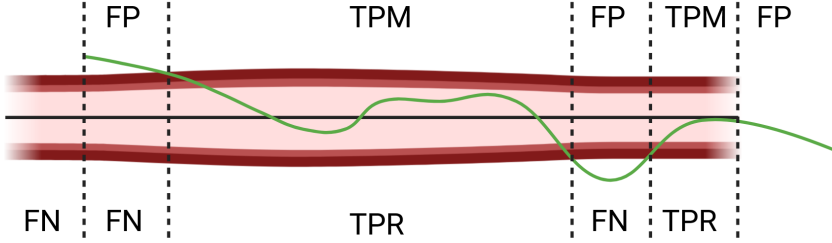


Figure 5.13: Schematic illustration of centerline points' classification into TPR, TPM, FP, and FN groups, inspired by [63]. Black line corresponds to the ground truth centerline G , and green line to the predicted centerline P . Created with [53].

Denoting $\|\cdot\|$ as number of point in given category, e.g. $\|FN\|$ - a number of false negatives, we define an overlap as:

$$OV = \frac{\|TPM\| + \|TPR\|}{\|TPM\| + \|TPR\| + \|FN\| + \|FP\|} \quad (5.24)$$

An **average inside (AI)** defined by Schaap et al. [63] is a measure that helps evaluate how accurate the tracking was in those parts of the coronary tree where the distance between the ground truth and predicted centerline is lower than the ground truth radius. In other words, it is calculated only for points classified as TPM, according to the description above. AI is then specified as an average distance between the ground truth and predicted centerlines for points fulfilling this condition.

Another two measures used in this project, as suggested by Radutoiu [13], are **coverage**: $C_{G \rightarrow P}$, that defines how much of the ground truth path has been covered by

the predicted one, and $C_{P \rightarrow G}$, which tells how much of the predicted centerline is reflected in ground truth data. Both of those measures use labels proposed by Schaap et al. [63] and are defined as:

$$C_{G \rightarrow P} = \frac{\|TPR\|}{\|TPR\| + \|FN\|} \quad (5.25)$$

$$C_{P \rightarrow G} = \frac{\|TPM\|}{\|TPM\| + \|FP\|} \quad (5.26)$$

$C_{G \rightarrow P}$ equal to 1 means that the whole coronary tree was captured by the tracing algorithm, and it is the most sought-after result. The lower the value is, the smaller part of the reference tree was found by the algorithm.

$C_{P \rightarrow G}$ equal to 1 means that the whole predicted path corresponds to the ground truth data and there are no regions that are not reflected in the reference centerlines. The lower the measure is, the more the algorithm misfits the ground truth data.

5.8.3 Curved Planar Reformatting

One of the methods of evaluating tracked centerlines is Straightened Curved Planar Reformatting (CPR) [64]. It is a method used for visualizing curved tubular structures, such as blood vessels, or, in this case, coronary arteries. In particular, it allows to visualize parts of the CCTA scan corresponding to a given centerline. It is especially useful considering that the ground truth segmentation of coronary arteries and ground truth centerlines are not of the highest quality and they do not cover the entire structure of coronary trees, making it difficult to evaluate quality of the tracer based on the ground truth data alone. The method relies on an extracted centerline to "straighten" coronary artery so that its entire structure could be seen in a single image. For each point on the curved centerline, a local coordinate system is defined, where z-axis corresponds to the tangent vector to the centerline, whereas x- and y-axes correspond to the cross-sections of the artery. Then, a transformation mapping from points on a curved centerline to points on a straight centerline is applied. CPR results in two 2D images, each corresponding to the middle cross-sections of the arteries. Height of the images corresponds to the length of the inspected centerline. An example of CPR visualization is shown in figure 5.14.

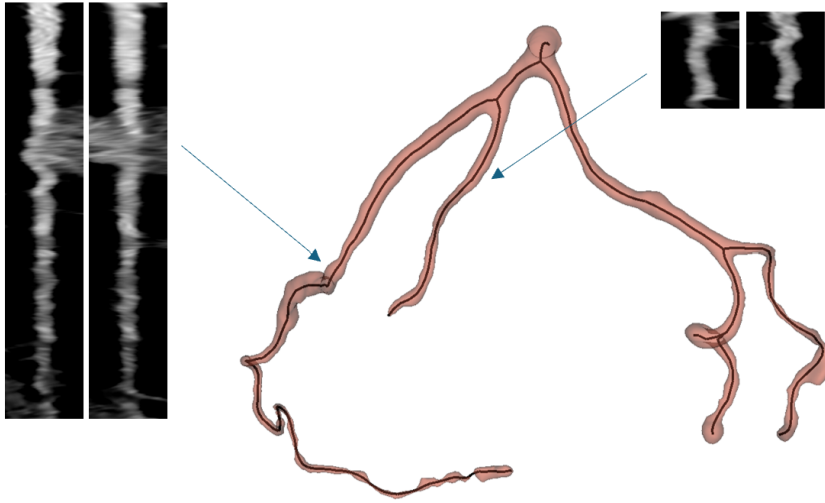


Figure 5.14: Example of CPR visualization for two chosen branches based on the ground truth centerlines. Heights of the CPR images correspond to the length of the chosen centerlines.

5.9 Phenotype clustering

As described in section 3.2, certain geometric features of the coronary arteries are linked to the increased risk of plaque formation. Having the centerlines of coronary trees, it is possible to calculate such features for each patient. Based on the set of the geometric features, it is interesting to see if phenotypes of coronary arteries can be detected. Phenotypes could then be analysed with fluid dynamics methods to see if they show characteristics associated with increased risk of plaque formation or rupture. In this project, we try to uncover the phenotypes in our dataset.

The analysis was based on the ground truth centerlines. For each coronary tree in our dataset, we extracted a fragment of LCA from the root until just after the first bifurcation point. Usually, there are two branches going out of the first branch point, corresponding to LAD and LCX. In cases where there were more paths branching out, we calculated the length of each path extending from the branch point and took into account only points belonging to the two longest ones, as they should correspond to LAD and LCX, being the longest branches in the coronary tree [26]. Figure 5.15 depicts how a fragment of interest is extracted from LCA.

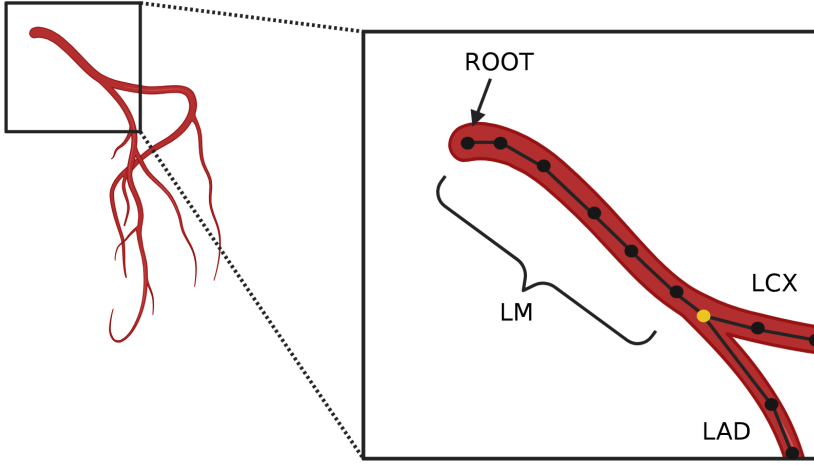


Figure 5.15: Visualization of the fragment extracted for the phenotypes clustering. Black line together with black dots correspond to the ground truth centerline and points on it. Yellow dot corresponds to the branch point. Created with [53].

The extracted fragment was used to calculate geometric features that are later used for clustering:

- Euclidean distance between the root point and the first point before the bifurcation point D_{LM} ,
- Geodesic distance between the root point and the first point before the bifurcation point L_{LM}
- Tortuosity of LM, defined as $\tau_{LM} = \frac{L_{LM}}{D_{LM}}$
- Radii of maximum inscribed spheres in points adjacent to the bifurcation point: φ_{LM} , φ_{LAD} , φ_{LCX} ,
- Angles between LM, LAD, and LCX: α_{LM-LAD} , α_{LM-LCX} , $\alpha_{LAD-LCX}$.

Schematic visualization of geometric features is shown in figure 5.16.

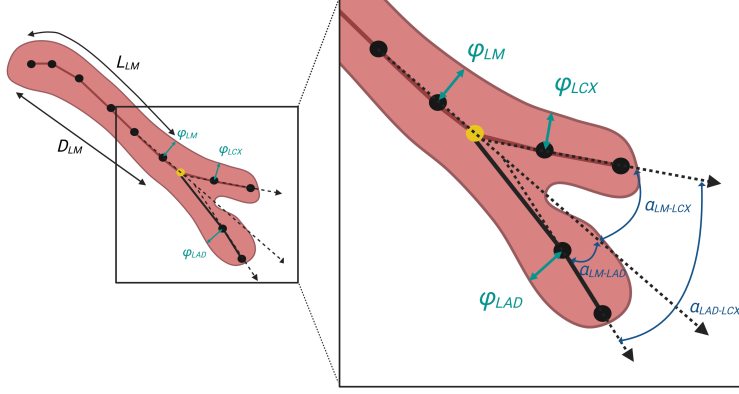


Figure 5.16: Geometric features used for LM clustering. Black dots denote points on the ground truth centerline, the yellow one to the branch point. Pink area corresponds to the segmentation of LCA. Created with [53].

Having extracted the features, the outliers were removed using a Z-score with threshold of ± 3 . Next, the data was standardized and agglomerative hierarchical clustering has been applied. As the linkage criterion for the clustering we used Ward's method [65]. Starting from singletons where each observation is a separate cluster, in each iteration two most similar clusters are merged all the way to the state where there is only one cluster left. The merging is performed based on the increment of total within-cluster variance being the sum of squared Euclidean distances. The merged clusters are the two that contribute to the lowest increase in variance. The Ward distance can therefore be formulated in the following manner:

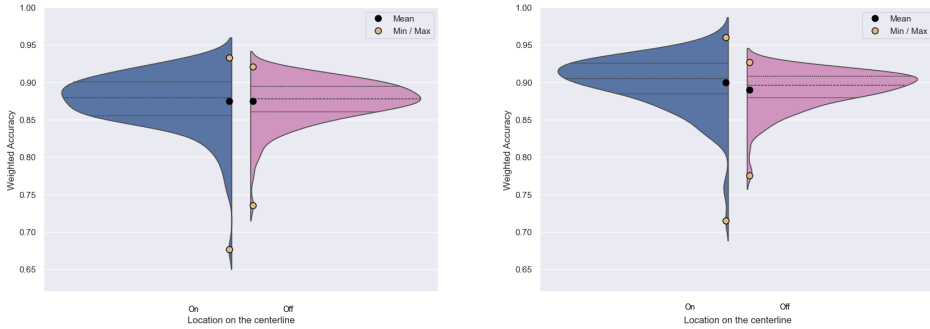
$$d_{\text{Ward}} = \sum_{i \in G \cup H} \|\mathbf{x}_i - \bar{\mathbf{x}}_{G \cup H}\|^2 - \sum_{i \in G} \|\mathbf{x}_i - \bar{\mathbf{x}}_G\|^2 - \sum_{i \in H} \|\mathbf{x}_i - \bar{\mathbf{x}}_H\|^2 = \sqrt{\frac{n_G n_H}{n_G + n_H}} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2 \quad (5.27)$$

where G and H indicate clusters, $\mathbf{x}_i$ represents feature values for i -th observation, $\bar{\mathbf{x}}_G$ and $\bar{\mathbf{x}}_H$ mean values for clusters G and H , respectively, n_G and n_H number of observations in clusters G and H .

The Ward's method tends to produce well balanced clusters and helps avoid chaining problem which can happen when using other methods like e.g. single linkage.

6.1 Centerline tracer

In order to evaluate the quality of the tracer networks, we performed two experiments on the test dataset, both for the regular tracer and for the reverse tracer. In the first experiment, we checked tracers' predictions for every point on the ground truth centerline, and calculated a weighted accuracy given the true best direction. In the second experiment, we drew ten offset points for each ground truth point on the centerline, and again, calculated a weighted accuracy for each of the points. For each of the test images, an average weighted accuracy was calculated for both experiments. The results are shown in figure 6.1.



(a) Regular tracer. For onset points, (min, mean, max) = (0.68, 0.87, 0.93). For offset points, (min, mean, max) = (0.74, 0.87, 0.92).

(b) Reverse tracer. For onset points, (min, mean, max) = (0.71, 0.90, 0.96). For offset points, (min, mean, max) = (0.78, 0.89, 0.93)

Figure 6.1: Comparison of distributions of tracer's weighted accuracies in test images. Left halves of violins show evaluation of points on the centerline, the right ones of offset points. Both violins have quartiles marked with dashed lines, as well as average value marked with black dot and minimum and maximum values marked with beige dots.

For the regular tracer, the average weighted accuracy is similar both for onset and offset points. In case of onset points, the distribution is a bit wider, with visibly lower minimum accuracy, and a bit higher maximum accuracy. For onset points, a true direction $\mathbf{d}_{\text{gt}}$ lies along the centerline. For offset points, $\mathbf{d}_{\text{gt}}$ is defined so that the distance to the centerline is decreased (eq. 5.5). It seems that a task of moving towards the centerline might be easier for the tracer than moving along it, thus the spread of the weighted accuracy distribution is lower for offset points. For the reverse tracer, the results are comparable, with slightly higher weighted accuracies

and narrower distributions than in case of the regular tracer. At the same time, it is important to note that offset points were sampled separately for both models, thus it is difficult to evaluate with certainty if the reverse tracer is indeed more accurate than the regular one.

Examples of tracer performance are shown in figure 6.2. The accuracies are lowest in areas of branch points and end points. In the left images for both tracers, there are branches that do not seem anatomically plausible - they are unusually short or follow the direction opposite to the rest of the coronary tree. In those cases, the tracer's performance is also lower. It suggests that having better quality data would result in even higher accuracies. The performance of both tracers is satisfactory, although visibly lower weighted accuracy values around end points can undermine the idea of using the reverse tracer to trace from ending points in order to fill the gaps around the branch points.

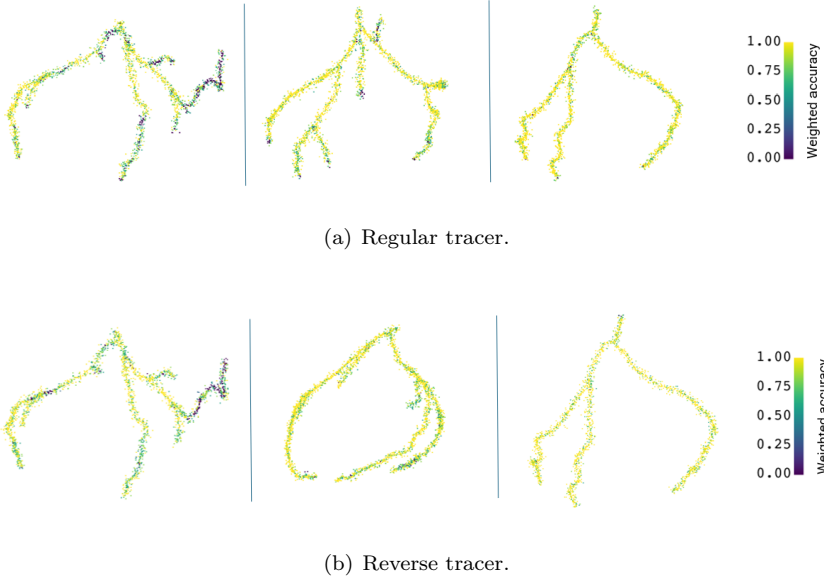


Figure 6.2: Comparison of tracers' performance on offset points for images with lowest, closest to average, and highest accuracy, from left to right.

6.2 Branch point detector

We evaluated the performance of branch point detector on images from test dataset. We have obtained the likelihood of being a branch point for all points laying on ground truth centerline and afterwards we followed the procedure explained in section 5.3 to

adjust locations of detected branch points based on their neighbourhood. Having gathered the predictions we evaluated them against the ground truth annotated branch points by calculating confusion matrix. We decided to classify the prediction as true positive if it lays within the range of 4.0 mm from the true bifurcation point.

Metric	Count
True Positives	611
False Positives	384
False Negatives	116

Metric	Value
Precision	0.614
Recall	0.840
F1-score	0.710

Table 6.1: Branch detector performance summary. Left, components of confusion matrix. Right, evaluation metrics: precision, recall and F1-score.

The results in table 6.1 show solid performance. The model was able to detect 84% of all true bifurcation points. At the same time, only around 61% of positive predictions were in fact correct. However, this is not that problematic for the tracing, as exploring areas of non-existent branch points is better than missing out on potential branch points completely. The most concerning part is 16% cases in which the branch detector actually failed to detect a branch. After further investigation, we noticed that among the images where the branch detector performs the poorest, many have incorrect segmentations. Examples of that can be seen in figure 6.3 where some of the cases classified as false negatives were not in fact true branch points even though the ground truth data suggested that. All in all, the model is performing adequately especially if we take into account that incorrect predictions are inflated because of the ground truth issues. We decided to accept it as a part of our tracing algorithm.

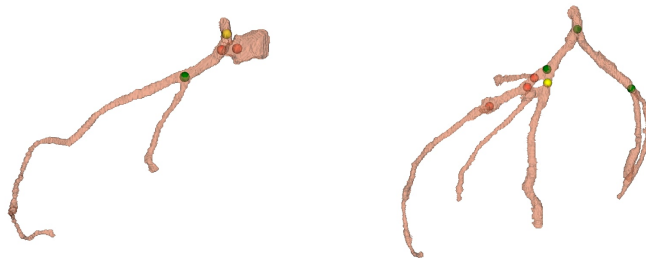


Figure 6.3: Examples of two of some of the poorest performance of branch detector. Green shows true positive branch points, yellow false positive and red false negative predictions. The segmentations are clearly anatomically implausible leading to incorrectly extracted ground truth bifurcation points.

6.3 Artery radius estimator

Radius estimator was evaluated by comparing its predictions for points located on ground truth centerlines from test dataset with corresponding true radii. For every point from the test set, we calculated the difference between estimated, r_{est} , and ground truth radius, r_{gt} , in a following way:

$$\Delta r = r_{\text{est}} - r_{\text{gt}} \quad (6.1)$$

It means that values larger than zero correspond to model overestimating true values, and smaller than zero to model's underestimation. Distribution of radii differences, as well as distributions of ground truth and estimated radii for the test set are shown in figure 6.4. The distribution of radii differences is centered around zero, meaning that for majority of examples radius estimator is very close to the true values. Looking at the distributions of ground truth and estimated radii, we can see that they are both right skewed with mode localized in a similar range. At the same time, low radii values are almost entirely not captured by the estimator, showcasing the flaws of the model.

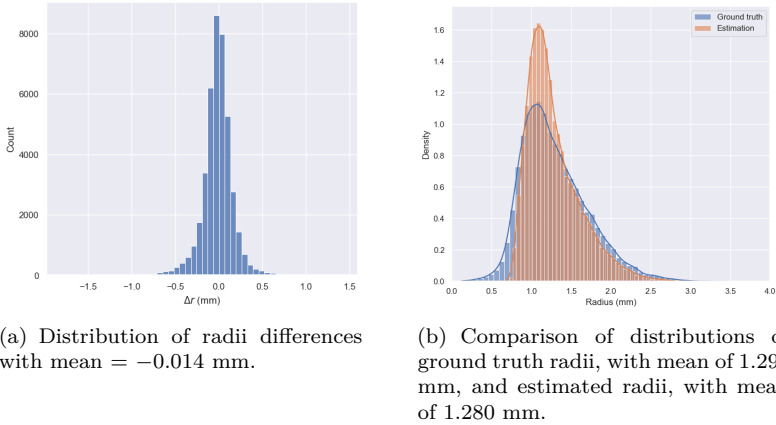


Figure 6.4: Evaluation of radius estimator on test images.

We calculated RMSE for each of the test images. The results are presented in figure 6.5. Even though the range of true radii values is between 0.12 and 3.48 mm, the model is quite accurate in its predictions, resulting in an average RMSE of 0.17.

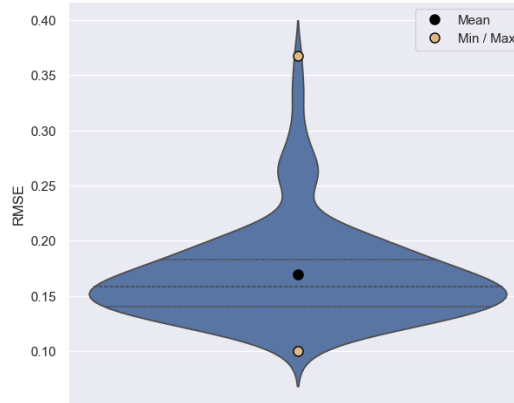


Figure 6.5: Violin of RMSEs of radius estimator evaluated on the test set images. (min, mean, max) = (0.10, 0.17, 0.37).

Examples of radii estimation performance are shown in figure 6.6. Image corresponding to the highest RMSE is the same one that was particularly problematic for the tracing networks, and it seems to be an outlier according to figure 6.5. It was mostly suffering from an underestimation issue, but no specific areas were particularly difficult for the estimator. At the same time, ground truth in this image can be corrupted, thus it might not be the best indicator of the model’s performance. In the other two examples, estimator’s performance is quite stable along the whole coronary tree. Model consistently overestimates radii at ending points of the coronary tree. Again, it might be related to the quality of the dataset.

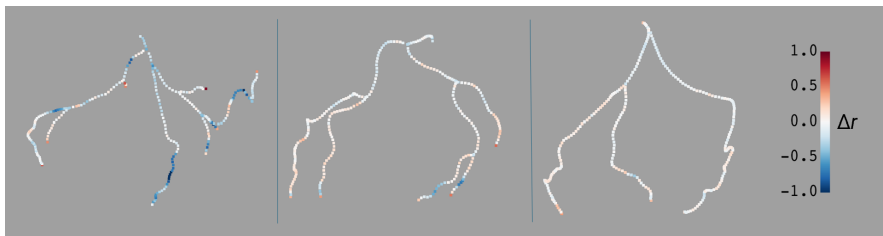


Figure 6.6: Examples of radii estimation for images corresponding to maximum, average, and minimum RMSE, from left to right, respectively. Points are color coded according to the definition from eq. 6.3. 6 points had values exceeding the thresholds shown in the legend, and thus were manually set to the limit values for the visualization purpose.

The performance of the estimator is very solid. Even though it does not fully capture true radii range values, it can definitely serve as a useful tool in branch spawning.

6.4 Artery likelihood estimator

We evaluated the performance of artery likelihood estimator on the test set images. For every point laying on the ground truth centerline we sampled 10 points offset by up to 4 mm so that both points inside and outside coronary artery are included. This resulted in a collection of 400 thousand points roughly equally distributed in and outside coronary artery. We obtained the prediction from the artery likelihood estimator for each such point and additionally calculated distance to the centerline.

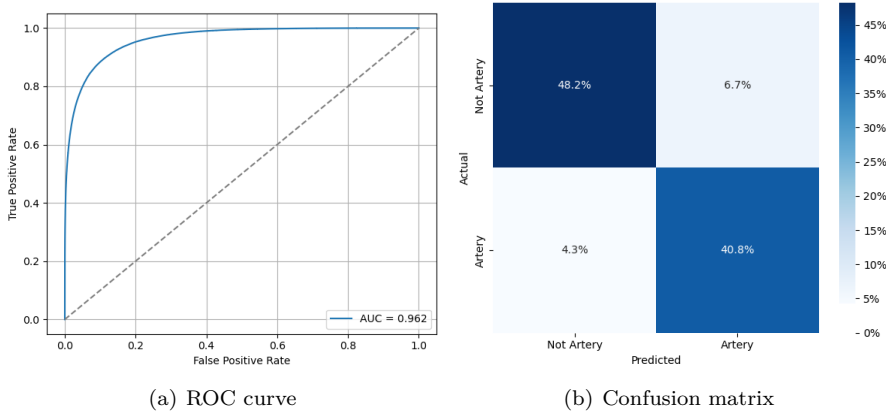


Figure 6.7: ROC curve and confusion matrix demonstrating performance of artery likelihood estimator on testset images. For the confusion matrix, classification threshold of 0.5 was used.

The model achieves accuracy of 0.89 and a high AUC score of 0.962. The ROC curve in figure 6.7 indicates that the the model is capable of distinguishing between the two classes well. Looking at the confusion matrix, in the 11% of cases where the model was mistaken it was more often a false positive rather than false negative. This goes well in hand with the purpose of using it for terminating conditions as it is less detrimental to not terminate the tracing early enough than it is to terminate it while actually being in the artery. The confusion matrix is based on the classification threshold of 0.5. However, in the terminating conditions implementation we are not using the binary predictions. Instead we evaluate a moving average of the likelihood with the threshold set to 0.25 which reduces the chance of getting a false negative further. Based on a single point prediction with the 0.25 threshold the model would produce a false negative prediction in 3.3% cases.

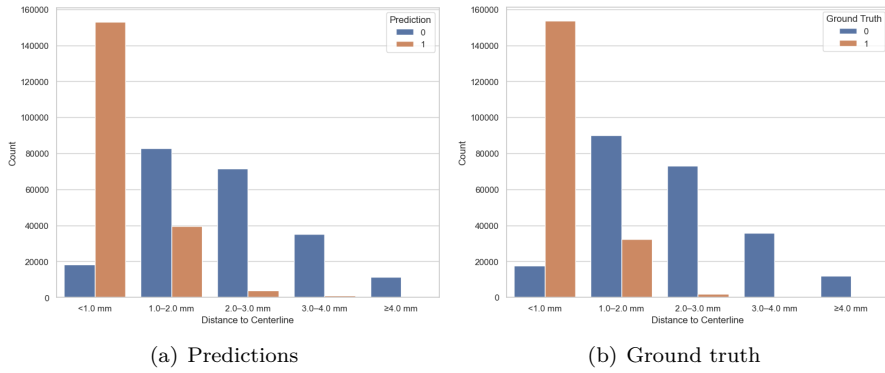


Figure 6.8: Histograms comparing relationship between predictions and ground truth labels with distance to the closest point on the centerline.

It is worth mentioning that both ROC curve and confusion matrix are based on the assumption that all misclassification cases are equal, however it is definitely more difficult to predict when the point lays very close to the artery surface compared to when it is far away from it. To make a closer look at how the estimator predictions change with the distance to the closest point on the centerline we grouped the points in 5 intervals of distances and investigated the distribution, as shown in figure 6.8(a). As expected, when moving away from the centerline percentage of positive predictions drop. After exceeding 2 mm, the model barely ever predicts that the point lays inside an artery. This is consistent with radius of arteries we observed in our data as shown in figure 6.4. Interestingly, there are cases of negative predictions even among the points laying less than 1 mm away from the centerline. Those are very often correct for points laying close to the narrow distal parts of the arteries. Additionally, in figure 6.8(b) we have shown a similar histogram created for ground truth labels. Histograms are almost identical showing that the model's predictions reflect the relationship with distance to the centerline well.

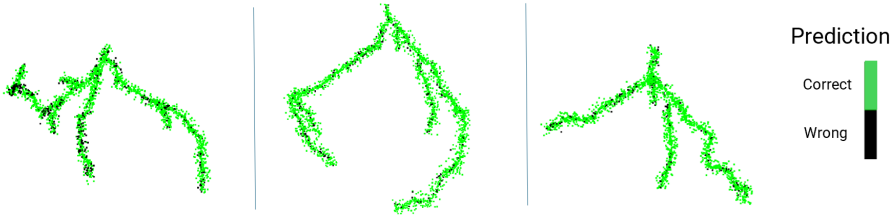


Figure 6.9: Panel showing artery estimator predictive performance on three test set examples. From left to right, images with lowest, closest to average, and highest accuracy are visualized. Points were sampled following the procedure described in section 6.4.

Naturally, some coronary arteries are more difficult to predict due to various factors such as tortuosity, the number of branch points or data quality. The accuracy does not however vary that much across images as it ranges from 0.845 to 0.921 for the middle 99% of observations. Therefore, even in the most difficult cases, model produces solid results. Figure 6.9 shows examples of the worst, average and best performance based on accuracy. There is no clear area in which the artery estimator underperforms. On the first artery we can see that the accuracy drops in tortuous region on the left, however, after closer qualitative investigation we noticed that there is an issue with the segmentation as it partly includes other anatomical structures. The right artery, with the highest accuracy proves that the model is capable of handling higher tortuosity. Even though this artery tree is fairly simple, the bottom right region becomes more twisted. The model predictions are still correct in this area. Notably, there is no indication that the model struggles in the bifurcation regions, suggesting that it can be well utilized in branch spawning.

The artery estimator is a robust model capable of indicating whether the state lays within the artery or not. It produces fully satisfactory results all around the artery tree. It can definitely aid the terminating conditions and determine the tracing direction in bifurcation points.

6.5 Coronary tree tracing algorithm

This section presents the results of the coronary tree tracing algorithm with different branch spawning techniques applied. The algorithms are first evaluated without the momentum procedure applied. Then, momentum is implemented for the most promising algorithm. Finally, performance of all techniques is compared to each other.

6.5.1 Direction-score-based branch spawning

Performance of direction-score-based branch spawning method has been evaluated on the test set using the metrics introduced in section 5.8. Visualization of the result is presented in figure 6.10. The method is characterized by a high $C_{G \rightarrow P}$ coverage with a mean of 0.835. $C_{P \rightarrow G}$ coverage average is equal to 0.819, indicating the algorithm has not suffered from too many false positives and most of the predicted centerlines were corresponding to the ground truth ones. Even though $C_{P \rightarrow G}$ is lower than $C_{G \rightarrow P}$, it is more beneficial to find as much of the true coronary tree at the cost of dealing with false positive, than the other way around. Overlap for the analysed method is equal to 0.812 on average. The evaluation metrics indicate the algorithm's strong ability to capture structures of coronary artery trees. There were 4 scans with $OV = 0$, and they were not included in the metrics' calculation.

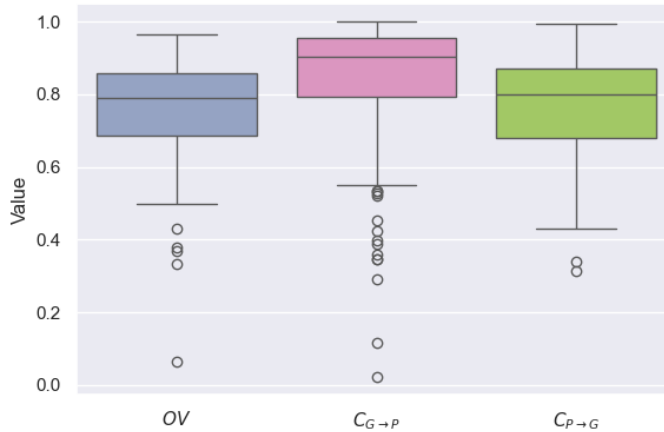


Figure 6.10: Boxplots of tracing evaluation measures. 4 images with $OV = 0$ were not included in the visualization.

Even though the evaluation metrics are satisfactory, a deeper look into centerline predictions revealed important issues. The direction-score-based branch spawning is an exploratory method where the tracer investigates multiple potential paths and, after many filtering iterations, returns a set of starting points of new branches. Despite the numerous filtering strategies and attempts to exclude any new starting points that result in similar paths, the algorithm suffers from an issue of multiple, similar paths originating from different spawn points of one branch point, that after a few steps unite into one centerline. Examples of limitations stemming from the exploratory nature of the method are shown in figure 6.11.

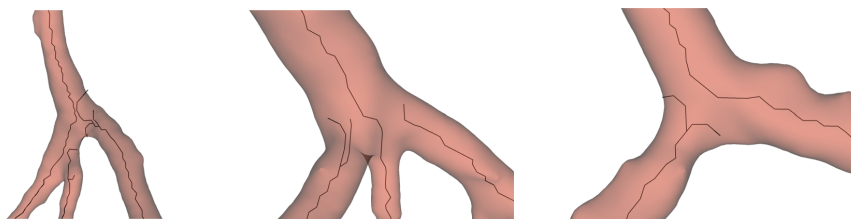


Figure 6.11: Issues due to the exploratory nature of direction-score-based branch spawning. Multiple paths originate from one branch point before they unite into one centerline.

6.5.2 Likelihood-sphere-based spawning

In this section we present the results of examining various approaches of creating a robust algorithm using likelihood spheres for designating the tracing direction at branch points.

The approach comprises two parts: creating a spherical direction likelihood map and indicating the tracing direction utilizing the von Mises-Fisher distribution as described in the section 5.6.2.

Figure 6.12 shows an example of several spheres built around the bifurcation points. It can be seen that they clearly indicate the areas of cross-sections with the artery segmentation. Even significantly thinner branches are still reflected in likelihood map. The qualitative assessment ensured us that the approach we employed creates a solid ground for denoting the tracing directions.

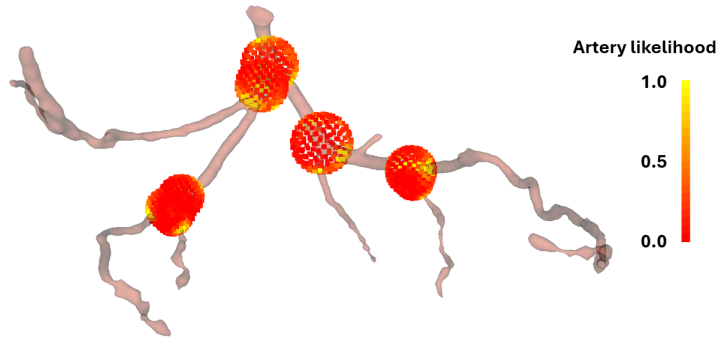


Figure 6.12: Example of artery likelihood spheres created around the bifurcation points of one of the testing images. The sphere correctly indicates how the artery is positioned, with higher likelihoods at the cross-sections with the segmentation.

6.5.2.1 Energy-based initialization

In order to test energy-based initialization described in section 5.6.2, we have taken an image with 5 ground truth branch points. For each of them, we have built a likelihood sphere and made 10 runs of energy-based initialization followed by fitting the mixture model and selecting the optimal one based on information criteria. The resulting optimal number of directions was compared against the true directions which we evaluated manually. Only a single image example was used for evaluation due to computational reasons and lack of ground truth number of splits in each branch point. The results have been summarized in the following way:

1. Missed - algorithm found less directions than expected,
2. Correct - algorithm found exactly the same number of directions as manually annotated,
3. Overestimated - found more directions than required.

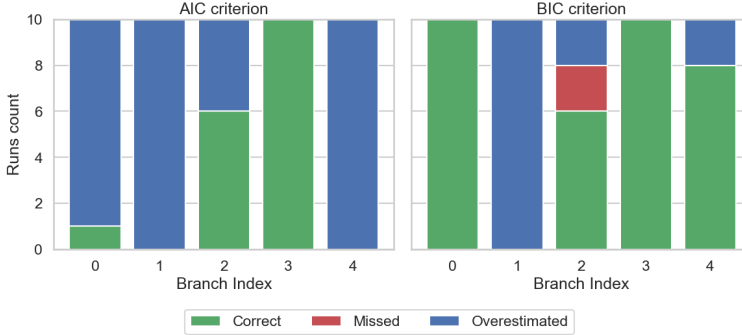


Figure 6.13: VMF model fitting with energy-based initialization ran 10 times for 5 branch points, optimized with AIC and BIC criteria. "Correct" means that the number of detected directions matched the ground truth; "Overestimated" means more directions were found; "Missed" means fewer were indicated.

The results of the experiment are shown in figure 6.13. Models selected based on AIC tend to overestimate the number of components more frequently than those preferred by BIC. This is in line with expectations, as BIC imposes higher penalty for model complexity. Given that the purpose of this model is to detect branch directions, fitting fewer components is far more detrimental than overestimating number of components. When overestimating, the tracer is required to follow one additional direction which will likely be quickly stopped by one of the terminating conditions, however, when not detecting one of the directions algorithm would simply miss out on whole potential branch and any of the subsequent branches. That being said, models optimized with both AIC and BIC perform well on average, and only in one case the model underestimated number of directions. Nonetheless, the biggest flaw of this approach is its dependency on a random seed. If the model produces different results on each run, it would require us to run this part of algorithm multiple times and select the most common number of components. This would in turn make the algorithm computationally inefficient and extend the time needed to trace the coronary tree.

6.5.2.2 KNN initialization

Having identified the issues behind energy-based initialization, we decided to pursue kNN-based approach. As previously stated, the model is deterministic, eliminating the randomness factor. The method requires tuning of two parameters, number of neighbours considered and relative threshold used to filter maxima. To find the most optimal parameters, we sampled a few images, constructed likelihood spheres for all ground truth branch points and, again, manually annotated the correct number of directions, producing a ground truth for comparing results from various models. We

tested the results on 83 spheres with four different configurations of parameters which were established to be most suitable after initial exploration. Analysed configurations of parameters are shown in table 6.2.

Model	Number of neighbours	Threshold
Model A	50	0.75
Model B	25	0.75
Model C	25	0.65
Model D	15	0.65

Table 6.2: Model configurations used in selecting the optimal kNN initialization technique.

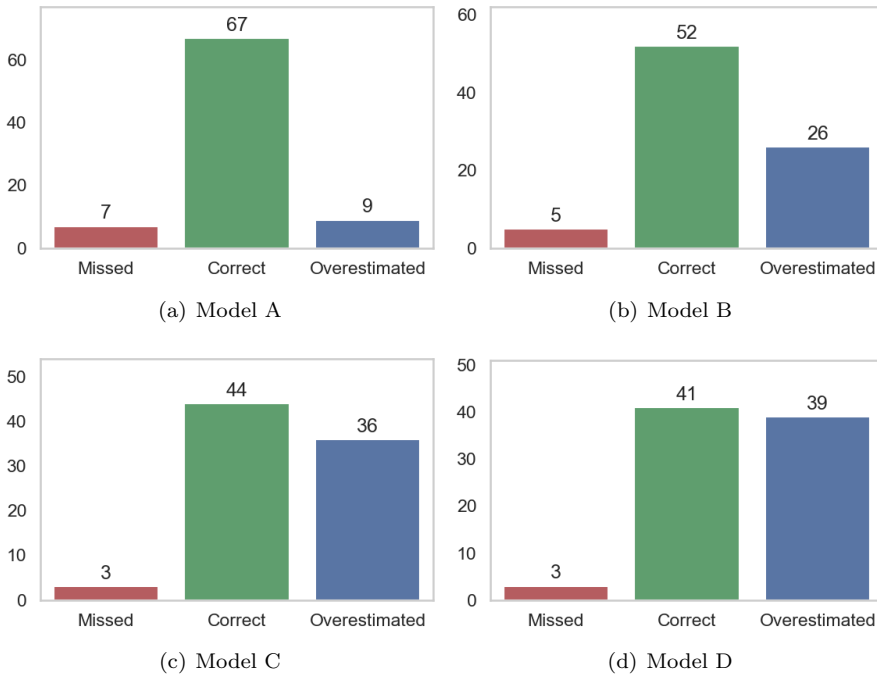


Figure 6.14: Directions found using different configurations of parameters in kNN initialization. “Correct” means the number of components matched the ground truth; “Overestimated” means more components were fitted; “Missed” means fewer were fitted.

Results produced by different models are shown in figure 6.14. Model A performs best in terms of number of “correct” hits, but it results in the highest number of missed directions, causing a risk of missing out on some parts of coronary artery tree. Since

the goal is to minimize the missed cases, more liberal model should be picked. Striking a compromise between limiting the number of misses and not inflating overestimated cases too much, we decided to select models B and C, and evaluate them together with the whole coronary tree tracing algorithm.

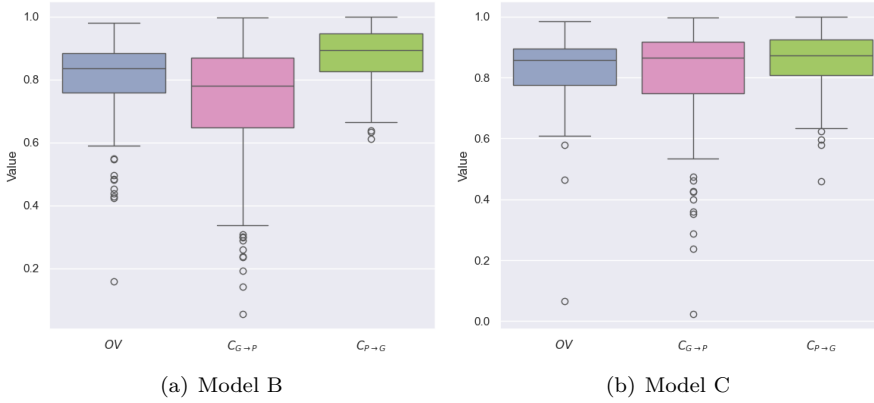


Figure 6.15: Boxplots showing performance of the tracing algorithm with two configurations of kNN initializations in likelihood-sphere-based spawning. Configurations are described in table 6.2. 4 scans with $OV = 0$ are not included.

Performance of tracing algorithms with Model B and Model C are shown in figure 6.15. Both models perform well, with relatively few outliers. Model B performs better in terms of $C_{P \rightarrow G}$ with the average of 0.876. However, when it comes to $C_{G \rightarrow P}$ and OV it is the Model C that stands out. For $C_{G \rightarrow P}$, the average values are 0.73 and 0.81 for Model B and C, respectively. Moreover, we can observe a higher interquartile range with Model B reaching 0.22 compared to 0.17 in Model C.

The evaluation metrics confirm insights from figure 6.14. Lowering the threshold for accepting the maxima translates into more tracing directions which, in turn allows, for detection of a greater portion of LCA. This comes at a cost of accepting that the tracer would sometimes detect false positive parts. Overall, the benefits outweigh the drawbacks, and Model C appears to be the strongest performer.

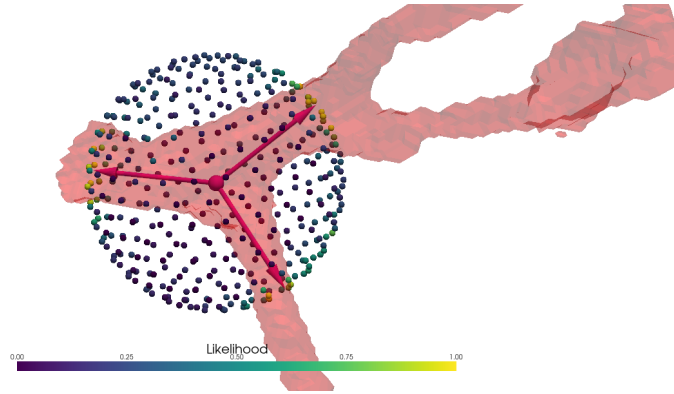


Figure 6.16: Example of directions found for one of the branch points using likelihood-sphere-based spawning approach. Sphere has been overlaid on the coronary artery segmentation. Colours on the sphere indicate cumulative artery likelihood in a given direction. Arrows show tracing directions predicted by the model.

Figure 6.16 shows the capabilities of detecting where the tracer should go using the likelihood-sphere-based approach. Likelihood sphere clearly indicates where the artery is located and shows higher likelihood at the cross-section with the segmentation. The von Mises-Fisher distribution fitting results effectively point towards those areas designating further tracing directions.

6.5.3 Tracing with momentum

As the algorithm incorporating likelihood-sphere-based branch spawning does not suffer from the issue of multiple, similar paths in branch point areas and is not a strictly explorative method, we decided to use it for evaluating the next algorithm improvements. As the next step, we evaluated the tracing algorithm with added momentum procedure. The boxplots of evaluation metrics are shown in figure 6.17.

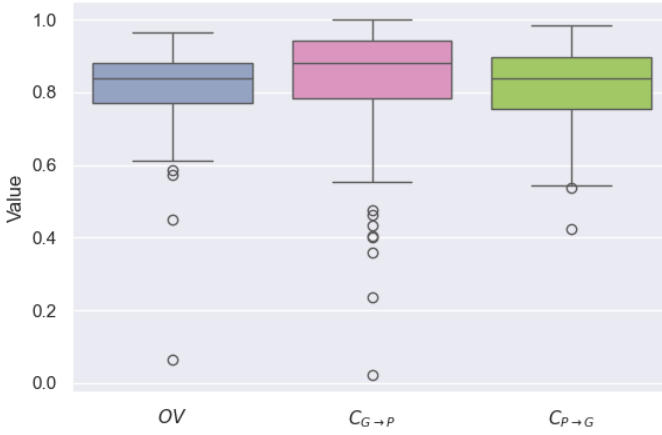


Figure 6.17: Boxplots of evaluation metrics after adding momentum procedure to tracing algorithm with likelihood-sphere-based branch spawning. 3 scans with $OV = 0$ were not included.

Mean $C_{G \rightarrow P}$ is equal to 0.837, bringing an improvement in the fraction of ground truth centerlines covered. $C_{P \rightarrow G}$ is slightly lower than before - the reason for that might be that the tracing proceeds further than the ground truth centerlines are extracted. Nonetheless, as the ground truth data is not ideal and the ground truth centerlines do not always cover the entire tree structure, the lower value of $C_{P \rightarrow G}$ does not necessarily mean a worse performance of the algorithm. Interestingly, one of the images that previously had $OV = 0$, was successfully traced after adding momentum, reaching $OV = 0.807$. Figure 6.18 showcases the examples of algorithm's performance after adding the momentum using CPR visualization method. It includes cases where tracing proceeds further than the ground truth segmentation, with CPR suggesting presence of coronary arteries. The CPR images' irregular shapes are most likely due to the lack of smoothing of the tracked centerlines, as the CPR is highly susceptible to the quality of the centerlines. The results suggest that the true performance of the algorithm might be higher than the evaluation metrics indicate, and having higher quality data could help determine the performance with better accuracy.

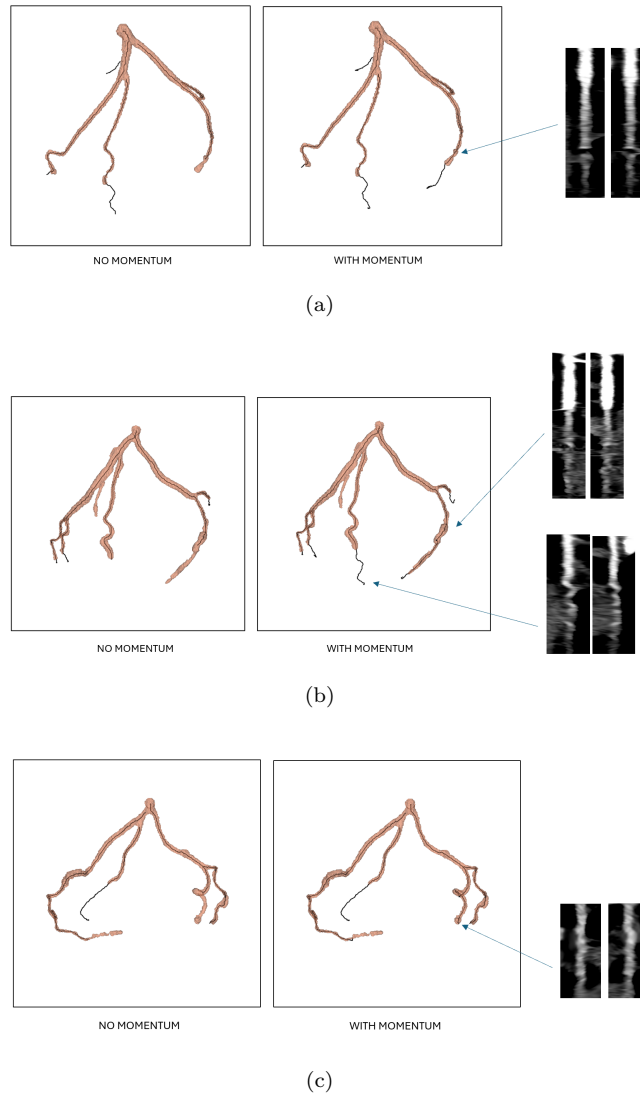


Figure 6.18: Evaluation of momentum procedure using CPR visualization method. In (a) and (b) there are centerlines traced with momentum that do not correspond to the ground truth segmentation, while CPR suggest presence of the coronary artery. In (b), one of the traced centerlines terminated too early without momentum applied, even though coronary arteries are visible both in ground truth segmentation and CPR visualization. (c) shows an example of momentum helping with high tortuosity branch.

6.5.4 Reverse tracing

In order to evaluate reverse tracing performance, we calculated the evaluation metrics for test set images. The reverse tracing was executed using ending points of centerlines traced using algorithm incorporating likelihood-sphere-based spawning and momentum. The results are shown in figure 6.19. Even though the $C_{P \rightarrow G}$ coverage has high values with an average of 0.863, meaning that most of the predicted centerlines correspond to the true ones, $C_{G \rightarrow P}$ performs much worse, with an average of 0.681, indicating that a large fraction of coronary trees was not captured in the predictions. The reason for such low values is that the reverse tracer, starting from ending points of previously found centerlines, was often going into a loop, unable to find the whole centerline. Examples of such issues are shown in figure 6.20.

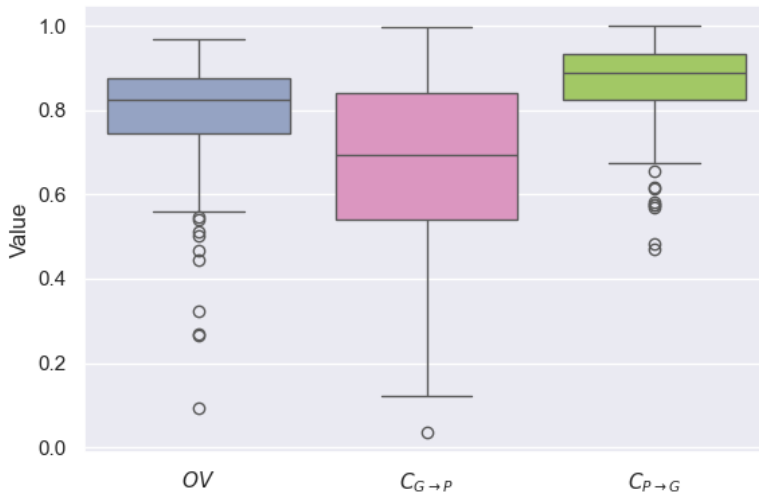


Figure 6.19: Boxplots of evaluation metrics for reverse tracing algorithm. 3 scans with $OV = 0$ were not included.



Figure 6.20: Example of reverse tracing algorithm performance. Image on the left shows the regular tracing algorithm with likelihood-sphere-branch spawning and momentum applied, image on the right shows reverse tracing. Green circles show areas in which reverse tracer succeeded in stitching up centerlines at branch points. Red circles indicate reverse tracing inability to capture some of the centerlines, resulting in short paths terminating too early, or completely missed branches.

Even though reverse tracing algorithm is successful in stitching up branches of the coronary trees, the fraction of centerlines that was not traced was too large to incorporate it in the final algorithm. The issues of the algorithm are especially pronounced for long centerlines, going beyond the ground truth segmentation. A possible solution for that would be to apply reverse tracing for instance from midpoints of centerlines tracked in a regular way, and then combining the results of a regular and reverse tracing together. Nevertheless, the reverse tracing algorithm in its proposed state is not functional as it misses to much of the ground truth coronary trees.

6.6 Final algorithm selection and evaluation

During the project we have experimented with multiple algorithm configurations, each tackling the problem of tracing the LCA from a slightly different angle. Within each configuration we tuned the parameters based on the performance on the test set. From all the tuned versions of algorithm configuration we decided to highlight 5 of them comparing their coverage, overlap and average inside metrics, and select the best performing one to use for deployment on CGPS data.

Model	$C_{P \rightarrow G}$	$C_{G \rightarrow P}$	OV	AI
DirectionScore	0.819 ± 0.112	0.835 ± 0.171	0.811 ± 0.123	0.113 ± 0.074
SphereStrict	0.876 ± 0.089	0.730 ± 0.198	0.802 ± 0.126	0.088 ± 0.038
Sphere	0.854 ± 0.098	0.810 ± 0.170	0.827 ± 0.110	0.095 ± 0.073
SphereMomentum	0.816 ± 0.106	0.837 ± 0.155	0.812 ± 0.109	0.097 ± 0.075
SphereReverse	0.863 ± 0.100	0.681 ± 0.211	0.787 ± 0.143	0.111 ± 0.159

Table 6.3: Performance metrics for different tracing algorithms (bold indicates best value in each column). Two highlighted rows correspond to two best performing models overall.

Table 6.3 shows comparison of chosen algorithms. **DirectionScore** denotes an algorithm with direction score spawning, whereas all models starting with **Sphere** incorporate likelihood-sphere-based spawning. Specifically, **SphereStrict** uses Model B from table 6.2 for initialization, **Sphere** - Model C, **SphereMomentum** - Model C and momentum, and **SphereReverse** - Model C, momentum, and reverse tracing. No algorithm is superior in every aspect as each approach has its strengths and weaknesses. The DirectionScore works well when it comes to finding where the artery lays but, at the same time, it sometimes tends to wander around, finding false positive parts of centerline. Judged by the metrics alone, the algorithm is robust, however, keeping in mind the issues coming from explorative nature shown in figure 6.11 we decided to discard it. Centerlines traced with SphereStrict are the closest to the ground truth when tracing inside an artery. At the same time, model does not produce many false positives as indicated by the $C_{P \rightarrow G}$, however, utilizing it means missing out on a lot of ground truth, as it only manages to find 73% of the whole tree. Similarly in SphereReverse, although 87% of the tracing is the actual ground truth, even less of the overall ground truth is found. We decided to discard both of them.

Following the principle that tracing too much is better than not tracing enough, we would rather choose an approach with $C_{G \rightarrow P}$ at the expense of $C_{P \rightarrow G}$. This led us to consider the Sphere and the SphereMomentum approaches as the two best algorithms. Both of them perform well in every aspect and the overall difference between them is subtle. We decided to investigate them more in depth.

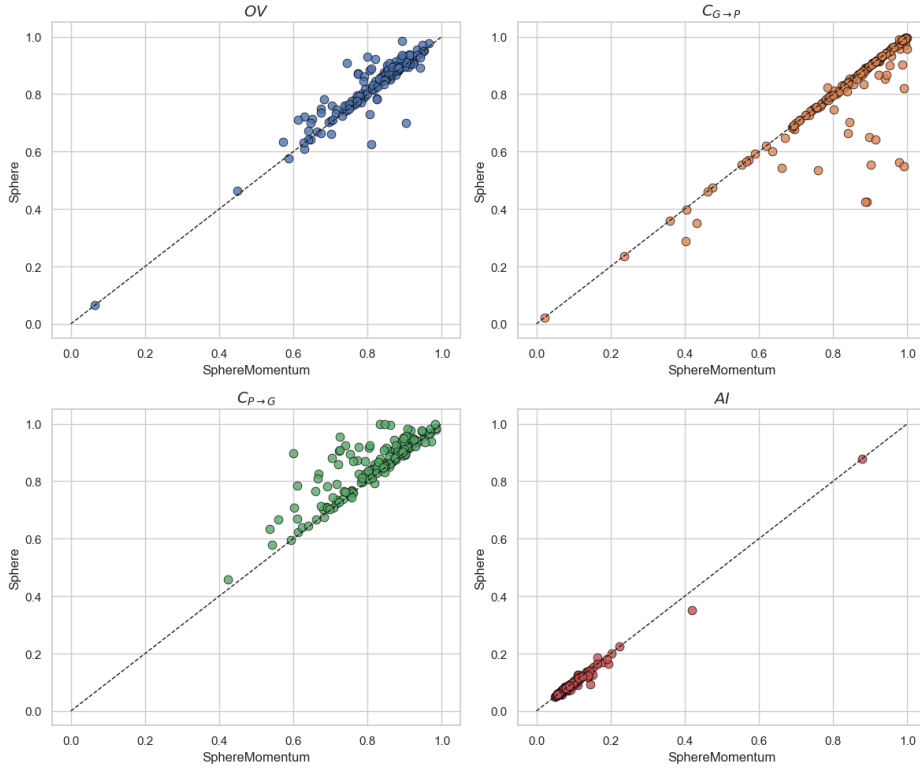


Figure 6.21: Scatter plots comparing Sphere and SphereMomentum algorithms. Each dot represents an image from the test set. The black line corresponds to a reference line indicating equal performance from both models.

Figure 6.21 allows to shed a new light on the metrics and compare the tracing performance on the image level. Most of the images lay on the reference line meaning that in the majority of cases the models perform equally well. In terms of AI, the differences for images deviating from the reference line are marginal and the two models' performance is almost identical. As for other metrics, it is clearly visible that model with momentum finds more of the ground truth at the cost of producing false positives. This alone could arguably tip the scales in favour of SphereMomentum. On top of that, one can see that there are several images in which SphereMomentum achieved almost 90% of $C_{G \rightarrow P}$, whereas Sphere reached only around 50%. In other words, for some images half of the whole tree is not detected in Sphere algorithm, while SphereMomentum detects almost the entire tree.

Examples of SphereMomentum performance are shown in figure 6.22. Predicted cen-

terlines have been visualized on top of the rendered heart volume. (A) showcases an image with the best overall performance, having $C_{G \rightarrow P} = 1.0$, $C_{P \rightarrow G} = 0.951$, and $OV = 0.964$. It means that the entire ground truth LCA was captured in the tracing and only very small parts exceed the ground truth segmentation of coronary arteries. (B) shows example of a scan with one of the highest $C_{G \rightarrow P}$ equal to 0.996. At the same time, $C_{P \rightarrow G} = 0.740$, which means that a significant part of the tracing does not correspond to the ground truth data. It resulted in $OV = 0.792$, which is lower than the average for the whole test set. The visualization indicates that, in reality, almost all predicted centerlines correspond to the coronary arteries visible in the scan, suggesting that the low $C_{P \rightarrow G}$ and OV values might be due to the segmentation quality, not the algorithm's performance itself. In (C), we can see LCA with high complexity structure and tortuous branches. In that case, $C_{G \rightarrow P} = 0.901$, $C_{P \rightarrow G} = 0.805$, and $OV = 0.824$. This example shows that the algorithm's performance is highly satisfactory in non trivial cases as well.

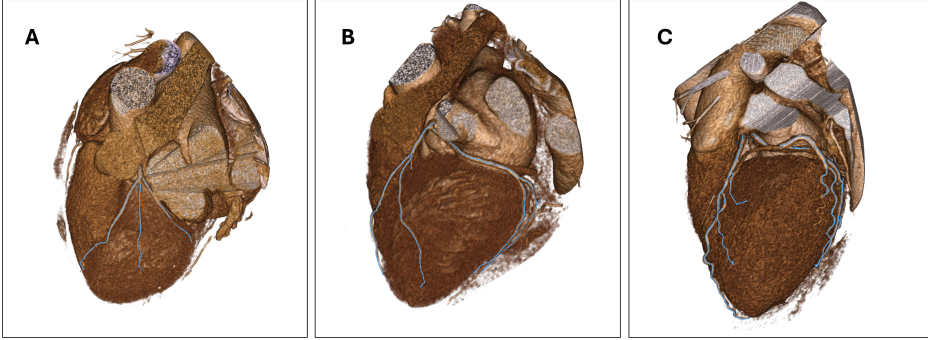


Figure 6.22: Centerlines (blue lines) predicted using SphereMomentum algorithm overlaid on the 3D rendering of a heart volume. (A) shows example with highest $C_{G \rightarrow P}$, (B) with high $C_{G \rightarrow P}$ and considerably lower $C_{P \rightarrow G}$, (C) shows algorithm's performance for a complicated tree structure with high tortuosity branches.

The visual assessment showed that the algorithm was able to find both LAD and LCX branches in 144 (89.44%) images in the test set, while in 8 (5%) only LAD was found, and in 5 (3.1%) only LCX. It managed to find at least one of these branches in 157 (97.52%) scans. For 3 images $OV = 0$.

All things considered, even though the performance of the analysed algorithms seems to be quite similar based on the evaluation metrics, after a closer look we decided that SphereMomentum slightly outperforms other configurations. It is able to capture the biggest fraction of ground truth LCA, and even though it does not have the highest $C_{P \rightarrow G}$ and OV , it is probably caused by the quality of the dataset. Keeping in mind the inaccuracies of evaluation metrics caused by data quality issues we supported our judgment with visual analysis, finally deciding to use SphereMomentum for the final

part of tracing algorithm analysis, i.e. evaluating its generalization abilities on a new dataset.

6.7 Copenhagen General Population Study

Due to the fact that the CGPS data is associated only with coronary arteries segmentation and starting points of LCA extracted as described in section 5.1, it was only possible to conduct a qualitative evaluation of the model's performance. To this end, we analysed visualizations of tracked centerlines on top of the LCA segmentations. We manually identified LAD and LCX branches for each scan, and decided to use the algorithm's ability to trace LAD and LCX as the main evaluation criterium. Out of 939 scans, both LAD and LCX were found in 728 (77.5%) images. In 48 (5.1%) scans only LAD was detected, and in 99 (10.5%) only LCX. In total, in 875 scans at least one of those branches was found (93%). In the rest of the images, LAD and LCX were traced only partially, thus they were not classified as successful tracing. Based on the visual inspection, for 722 (76.95%) scans almost entire coronary tree was traced, meaning that the centerlines covered either the entire segmentation, or only some very small subbranches were missed. For 12 images $OV = 0$, meaning that none of the coronary tree was traced. In 42 scans traced centerline was very short, reaching only the first bifurcation point or terminating even earlier.

Figure 6.23 shows examples of successful (in terms of finding LAD and LCX) tracing on top of the LCA segmentation. It can be seen that the algorithm is capable of tracing the LCA both for relatively simple tree structures (top row), as well as more complicated ones (bottom row). Its performance is highly satisfactory for branches of high tortuosity, showing that the model can capture even most complex branches. In some cases, more branches were traced than the segmentation indicates (image id: 0113). At the same time, the very beginnings of those branches are included in the segmentation, suggesting that algorithm's predictions might be correct, and the issue lies in the quality of the dataset. For most of the scans were tracing of LAD and LCX was successful, predicted centerlines terminated far away from the end of the segmentation. Due to the limited access to the scans themselves, it was difficult to assess if this is due to incorrect segmentation, or if the terminating conditions formulated based on ImageCAS do not adapt well to the new dataset and some improvements would be required. Nevertheless, it is once again worth noting that tracing too far away is less harmful than stopping too early.

Overall, the algorithm proposed in this project seems to be capable of generalizing well to the new dataset. A more in-depth assessment including raw 3D scans could reveal more about the algorithm's true performance. The analysis also lacks a quantitative evaluation, which was not possible due to lack of ground truth centerlines extraction. Nonetheless, the visual inspection shows that the results are promising and the algorithm can be successfully used on data acquired with a different scanner compared to one used in ImageCAS data, on which the method was trained.

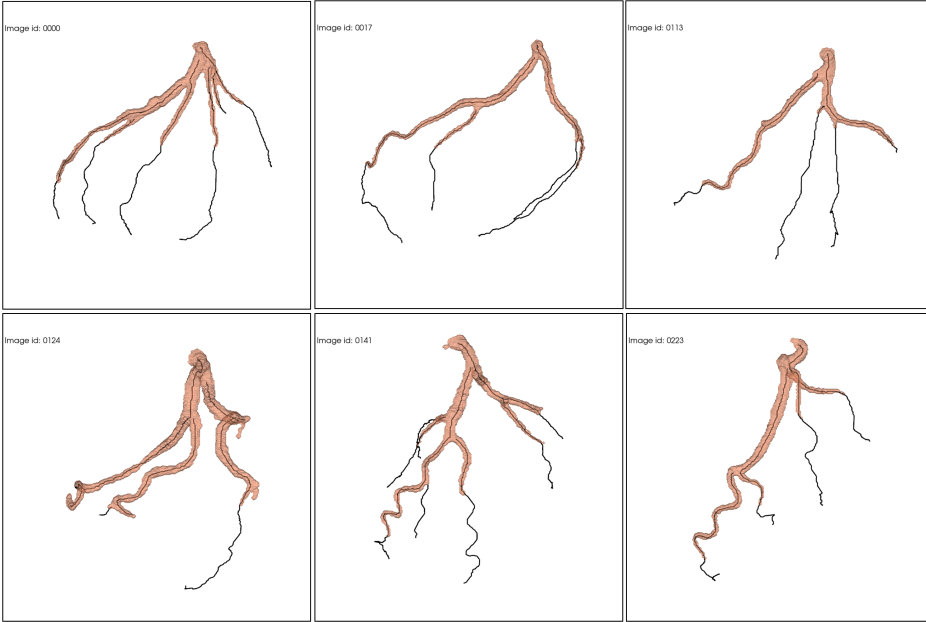


Figure 6.23: Examples of tracing on CGPS dataset. In black, traced centerlines are showed. In pink - ground truth segmentations of LCA. Top row shows examples of coronary trees with relatively simple structure, whereas bottom row includes examples of tree with more subbranches and higher tortuosity.

6.8 Phenotype clustering

As the secondary aim of this project, we performed an analysis aimed at discovering morphological phenotypes based on LMCA fragments of coronary arteries of patients included in ImageCAS dataset.

After cleaning the dataset and removing outliers, our sample consisted of 693 samples. As described in section 5.9, for each sample 9 features related to length of LM, radius and branch angles were calculated and a hierarchical clustering using Ward's method has been performed.

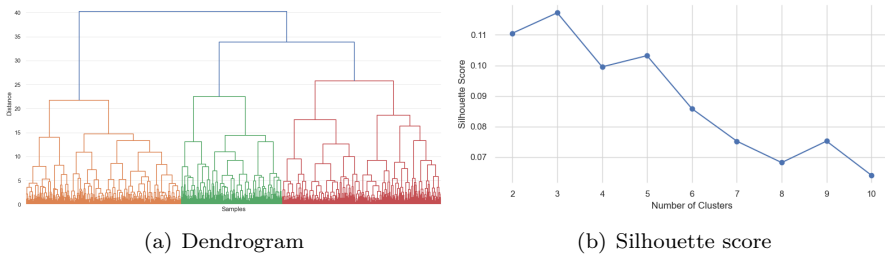


Figure 6.24: On the left, dendrogram showing division of samples between hierarchical clusters. Distance between each clusters has been shown on y-axis. On the right, silhouette score for different numbers of clusters.

Results in figure 6.24 indicate that the optimal number of clusters lies somewhere between 2 and 4. Following the silhouette score and exploration of results we decided to split our samples among 3 phenotype clusters. Wang et al. [17] discovered 4 phenotype groups which arguably could also be chosen in our case, however, after deeper investigation we noticed that the additional split of one of the clusters does not seem to produce more diverse results in our dataset. The 3 cluster split gives a fairly equally distributed phenotype groups which can be observed on the dendrogram in 6.24 by the division of samples along the x-axis.

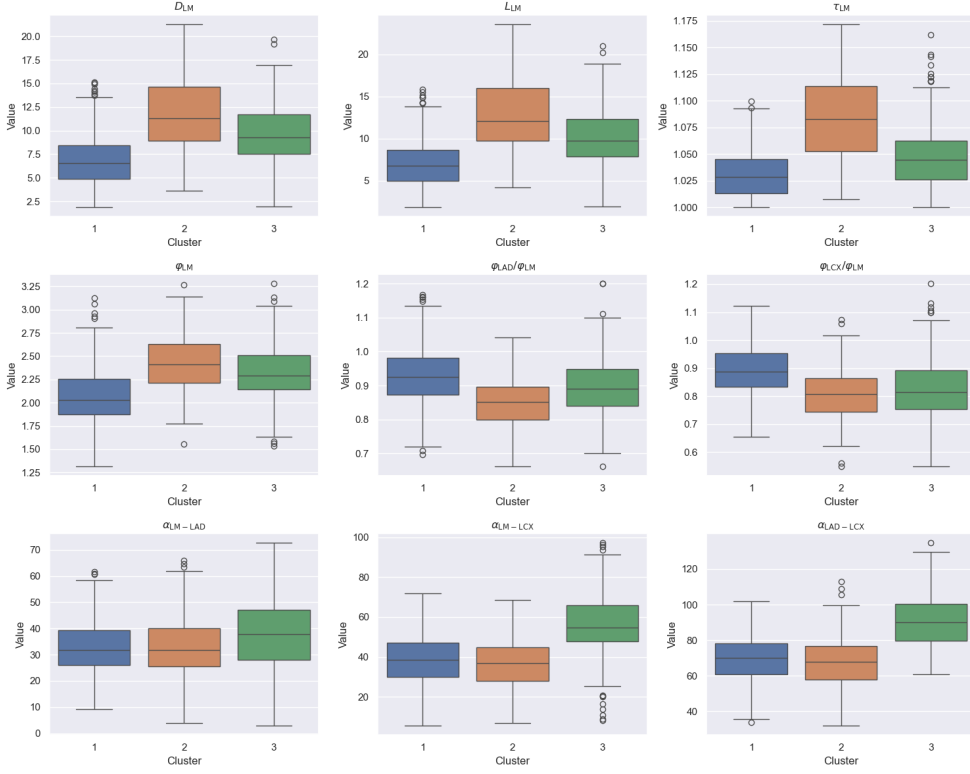
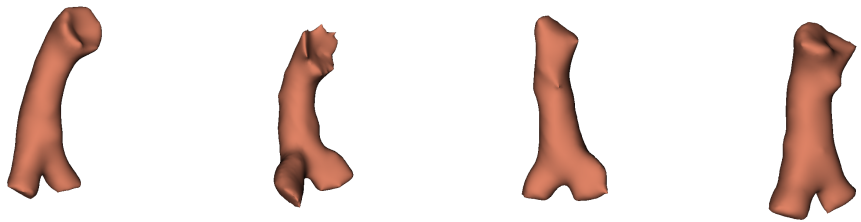


Figure 6.25: Box-plots of all features calculated separately for samples in each cluster.

Phenotype groups have their unique characteristics as indicated by figure 6.25. Second (orange) cluster includes samples with the longest Euclidean and geodesic distances along with the highest tortuosity. On the other end, the first cluster (blue) consists of patients with the shortest and least tortuous LMs, at the same time having the highest ratios of radii. The third cluster's (green) uniqueness lies mainly in angles between the two branches and between each of the branch and LM as those patients tend to have the most spread out branches.



(a) Phenotype 1



(b) Phenotype 2



(c) Phenotype 3

Figure 6.26: LM fragments for most representative samples of each phenotype found by minimal distance to averages across all features.

The findings from quantitative analysis of clusters are also reflected in figure 6.26. The presented samples are the most representative for their respective phenotypes having the minimal distance to the cluster average across all features.

Observations within each phenotype group appear to have distinctive traits related to their shape and size. Phenotype 1 cases are short and have branches almost as thick as the LMs. Those presented in figure 6.26 have both euclidean and geodesic distances of around 7 mm and both $\varphi_{\text{LAD}}/\varphi_{\text{LM}}$ and $\varphi_{\text{LCX}}/\varphi_{\text{LM}}$ of around 0.9. Phenotype 2 is clearly elongated and curved. Presented samples measure from 11 to 13 mm and have the highest tortuosity of above 1.08. Phenotype 3 patients stand out with high values of $\alpha_{\text{LAD-LCX}}$. The angle for the samples presented in the figure measures around 90 degrees but for some cases within the cluster can reach even to 120 degrees.

Even with relatively few features, patients' LMs can be divided into groups of unique characteristics. Those unique characteristics are the ones that were found to be associated to certain health risks. Short LMs, represented by phenotype 1, are often associated with CAD. What is more, short LMs are linked to higher risk of proximal plaque formation, increasing the severity of a disease [37]. High tortuosity arteries (phenotype 2) are of increased risk of developing plaques. According to research, segments of LCA with plaque were on average 30.8% more tortuous than others [38]. As for phenotype 3 with the largest bifurcation angles, they are attributed to the lower risk of proximal plaque formation in LAD and LCX branches [39]. In the research, small branch angles were linked to high risk of proximal atherosclerotic disease in LAD and LCX, with stronger correlation in LCX.

Our results show that by calculating few relatively simple metrics based on the center-line and radius of the coronary artery we can divide patients into phenotype groups and mark those with an increased risk of forming a plaque. Together with the artery tracer, this could serve as a tool to support various types of research aimed at understanding the risk of cardiovascular diseases, or as a solution for use in clinical settings to automatically identify patients at risk of developing stenoses.

7.1 Centerline tracer

The centerline tracer’s goal is to predict a direction $\mathbf{d}$ in which the tracer should follow being placed in a current location. It is guided by a direction score quality function (eq. 5.5) that encourages the tracer to follow the centerline in a direction of a blood flow, while staying close to it. The choice of model architecture has already been proven effective in previous works [13], with the size of the input image patch of 41 x 41 x 41 voxels being large enough to cover the entire coronary artery cross-section and its immediate surroundings.

The centerline tracer network achieved an average weighted accuracy of 0.87 both for onset and offset points (figure 6.1), manifesting its strong performance. As the weighted accuracy measure is based on a cosine similarity, it takes into account not only the correct predictions, but also close matches. Thus, its high value might be influenced by having a lot of predictions close to the true directions, but not equal. The distribution of weighted accuracies for both onset and offset points is similar in shape, but quite wider for the onset points. As the direction score function shows (eq. 5.5), the focus for offset points is mostly put on reducing their distance to the centerline, while for onset points, apart from staying close to centerline, the new locations have to follow the centerline in the blood flow direction. It suggests that the task of staying close to the centerline might be slightly easier than following it in the specified direction.

Visual inspection of centerline tracer (figure 6.2) allowed to capture its limitations. Branch points were one of the areas where the weighted accuracies were visibly lower than in the other parts of the coronary tree. It is in line with intuition, as in the branch points there are two (or more) paths to follow, and choosing the one that does not correspond to ground truth results in low weighted accuracy, even though the tracer would still follow the centerline correctly. The tracer’s performance is also highly influenced by data quality. As shown in figure 6.2, an example of coronary tree with lowest weighted accuracy suffers from anatomical implausibility, where one of the branches makes a sudden turn upwards, showcasing problems with ground truth segmentation. Points on the mentioned branch have very poor accuracy, deflating the overall score achieved for the test set.

The reverse centerline tracer displays comparable results to the regular one, and is suffering from the same limitations.

7.2 Branch point detector

The branch detector model is responsible for identifying branch points on the already traced centerline of the coronary artery tree. The implemented version is capable of identifying bifurcation points with the recall of 0.84 (table 6.1). The true performance of the model is difficult to evaluate accurately due to segmentation data quality issues often leading to incorrect positioning of ground truth bifurcations (figure 6.3). This is, in turn, reflected by the inflated false negative count. The model tends to frequently produce false positive predictions with only around 61% of its positive predictions being correct. This, however, is not troublesome as it most often means additional computational time of tracing in quickly terminated areas. Beyond the quantitative assessment, we observed in our qualitative evaluation that the room for improvement is not necessarily in the ability to detect that there is a branch point in a particular area, but rather the exact positioning of the bifurcation point. All in all, the model is adequate which is also reflected by the results of the whole tracing algorithm.

7.3 Artery radius estimator

Artery radius estimator is a model predicting a radius of coronary artery at a given location. Its performance is very strong, with the distribution of differences between true and estimated radii being centred around zero (figure 6.4). The distribution of predicted radii is similar to the distribution of the ground truth, but it is clear that the model has issues with correctly estimating the lowest values of up to around 0.75 mm. The input patch, similarly as in previous models, has size of 41 x 41 x 41 voxels, corresponding to 20.5 x 20.5 x 20.5 mm in physical dimensions. It means that the parts of coronary arteries with lowest radii correspond to relatively small areas of the image patch, making it difficult for the model to detect appropriate patterns in the scans. Experimenting with different input sizes, or creating an ensemble model combining a few networks with different input sizes, could help guide the solution to capture different level of details in the scans and produce better results.

An important limitation of the radius estimator is the data it has been trained on. As described in section 5.1, the ground truth radii correspond to radii of maximum inscribed spheres centred on each point of the centerline. The maximum inscribed spheres were created based on the dilated segmentation, as the algorithm used for centerline extraction (and thus, the spheres) was not accurate enough in very narrow regions of arteries. It means that the ground truth data does not correspond precisely to the true artery morphology, making the solution too faulty for the potential clinical use. At the same time, since in this project radius estimator is mainly used for finding starting points of branches in the coronary tree, its performance is highly satisfactory, regardless of the ground truth data.

7.4 Artery likelihood estimator

The artery estimator is a model tasked to predict the likelihood of a point located in 3D image space laying inside the coronary artery. The estimator is utilized in tracer's terminating conditions and in branch spawning to indicate the tracing directions at bifurcation points.

The model achieved an impressive performance with the accuracy of 0.89 and AUC score of 0.962 as shown in figure 6.7. The model showed strong capability of indicating when the tracer state is no longer inside the artery. With the 11% misclassification cases there is an advantage of false positive over false negative cases with the latter happening for only 4.3% predictions. From the standpoint of terminating conditions, falsely predicting that the state is within the artery is less harmful than incorrectly predicting it to be outside while it is, in fact, still inside. The former would cause tracer to wander away from the artery, in worst case scenario resulting in falsely tracked parts of the artery tree. The latter, on the other hand, could cause the tracer to terminate too early, potentially missing out on parts of the true artery. It is worth mentioning that the above results are based on the predictive threshold of 0.5. In the final implementation of the estimator in terminating conditions we are using the likelihood itself, not the binary prediction, to evaluate whether the tracer should stop. Furthermore, the condition is evaluated based on the average of last three steps and the average likelihood needs to fall below 0.25 to enforce stopping of the tracer. This way the algorithm is even less likely to be terminated too soon.

Figure 6.8 shows that there is a strong relationship between predictions and distance to the centerline. The histograms created for both ground truth labels and predicted ones are almost identical, showing how well the estimator handles predictions on the test dataset. The models outputs mostly positive predictions for distance up to 2 mm to the closest point on the centerline, making it consistent with the results from figure 6.4 and human anatomy [66]. The negative predictions tend to happen close to the centerline, however, this can be attributed to the most distal segments of the coronary tree where the artery becomes thinner. It can also be attributed to the fact that offset points sampled from the end points, while laying close to the centerline, were outside of the LCA.

Finally, the model performs consistently across all the images included in the test dataset with the accuracy of 99% central observations contained within 0.845 and 0.921. It is also consistent across the entire coronary artery tree, showing no difficulties at branch points or in the more distal segments. The results showed no concerns regarding the suitability for the respective use cases within the algorithm. It serves as a highly appropriate model for both likelihood-sphere-based spawning and terminating conditions.

7.5 Coronary tree tracing algorithm

7.5.1 Direction-score-based branch spawning

Direction-score-based branch spawning is an explorative method that compares multiple paths starting in a detected branch point based on the accumulated direction quality score. To be precise, we explore 50 potential paths in the directions defined by a Fibonacci sphere.

In ImageCAS, the algorithm managed to capture a vast majority of the ground truth centerlines, as suggested by $C_{G \rightarrow P}$ (figure 6.10). It is important to remember that the branch spawning procedures should not be evaluated in isolation from the branch detector performance. Any false negative produced by branch detector harms the final performance score, as it makes it impossible to find any new paths in the false negative area. What is more, we sometimes observed that the predicted branch points were offset from the true location towards one of the branches. As the set of exploring directions is created based on the branch point location, the quality and presence of the new paths is highly influenced by it.

The results show high coverage and overlap values. However, the method is based on a wide set of heuristics, including specific rules for filtering path candidates. This poses a risk of the solution potential inability to generalize to new datasets, and puts into question possibility of further usage in the clinical or research setup. Even though the set of heuristics was developed on ImageCAS data, in many cases it was still insufficient for filtering out the path candidates (figure 6.11). For multiple scans it caused issues with having tracked multiple paths around the branch points, that were often quite offset from the middle of the coronary artery (sometimes even going beyond the artery segmentation), and only after a while they united into one centerline.

Taking it all into account, the algorithm using direction-score-based branch spawning seems to produce satisfactory results on ImageCAS, but it suffers from multiple issues related to its explorative nature and heuristic-based approach, showing the need for a more robust and flexible algorithm.

7.5.2 Likelihood-sphere-based branch spawning

The second branch spawning method we explored was based on the likelihood sphere created using the artery likelihood estimator model. This approach creates a 3D sphere with the artery likelihood heatmap around the bifurcation point and fits the vMF model to indicate the tracing directions.

We confirmed visually that the likelihood sphere built around the branch points reflects the artery segmentation making it suitable for indicating the directions (figure 6.12). The task of establishing tracing directions based on the created likelihood sphere was solved using a mixture of von Mises-Fisher distributions (eq. 5.10), suit-

able for 3D spherical likelihood, which we decided to fit using least-squares numerical optimization.

The k-Nearest Neighbors approach was the final method used for initialization. Other approaches, such as random initialization, or energy-based initialization, suffered from varied quality of outputs. Apart from making the results deterministic, kNN resolved the problem of establishing how many directions should be found. This technique requires the tuning of number of neighbours and local maxima acceptance threshold. We tested a few combinations described in table 6.2 to finally choose Model C configuration with 25 neighbours and threshold of 0.65 striking a compromise between missed and overestimated directions.

With the implementation of this configuration the final tracing algorithm achieved an OV of 0.827 and a $C_{G \rightarrow P}$ of 0.81. The model was also consistent among all the test images as indicated by figure 6.15. After qualitative assessment we found out that the likelihood-sphere-based method rarely misses a tracing direction. The cases with missing directions were mostly associated with segmentation issues. Although there were very few cases where the spawning itself was underperforming, we noticed that the method relies on the branch detector's ability to indicate the exact placement of the bifurcation point. In some cases where the predicted branch point is offset towards one of the branches, the created spherical likelihood map does not reflect the branching structure well enough for the method to correctly indicate directions. It should also be emphasized that although all the networks have been trained using training and validation datasets, some heuristics of the algorithm were tuned on the testing dataset possibly making the performance subtly biased.

A very frequent reason for tracing termination was the tracer's oscillations. Thus, we implemented a momentum procedure along with the most favourable configuration of the algorithm - Sphere (table 6.3). The momentum helps guide the tracer out of the problematic areas. It increased $C_{G \rightarrow P}$ from 0.810 to 0.837, helping to cover larger parts of ground true centerlines. Even though $C_{P \rightarrow G}$ decreased from 0.854 to 0.816 due to the fact that tracing was often terminating further than the ground truth segmentation, CPR visualizations showed presence of the coronary arteries (figure 6.18) in the alleged false positive. It suggests that the performance of the algorithm might be better than demonstrated by the evaluation metrics. The momentum algorithm is based on a direction between a point visited right before entering the loop, and a point visited 5 steps earlier. The characteristics of this path segment, such as tortuosity or curvature, are not taken into account. It means that, when the oscillations happen in the location where the artery bends, a calculated direction might be not accurate enough, causing the tracer to leave the artery region and terminate due to low artery likelihood. A more robust way of calculating the momentum, taking the aforementioned features into account, could potentially make the algorithm even more efficient.

The design of the coronary tracing algorithm suffers from one significant limitation: there are empty gaps between detected branch points and starting points of new

paths. To solve this issue, we tried implementing a reverse tracing algorithm, in which the reverse tracer would start from the end points of branches found with a regular tracing, and proceeded in the direction opposite to blood flow. Unfortunately, that resulted in entire branches not being covered due to the tracer terminating very early (figure 6.20). At the same time, for branches that the algorithm managed to cover, it showed its excellent abilities in stitching up previously disconnected parts of LCA. Nevertheless, the reverse tracing is not adequate enough in its current state. A potential improvement would be to modify the reverse tracing so that it starts its course e.g. from midpoints of each branch and combine the results of regular and reverse tracing together. Another idea would be to apply an adjusted version of momentum to aid the tracer in the troublesome areas. Resolving this issue was not crucial for the purposes of this project but should this algorithm be employed in e.g. clinical setup it would need to be addressed.

A possible addition to the solution proposed in this project would be to apply appropriate post-processing to smooth out the tracked centerlines. The outputs of the algorithm in their raw form wander a bit around the true centerlines, influencing the results. An example of that can be seen in CPR visualizations in figure 6.18. Any further research using the method proposed in this project could arguably benefit from this additional post-processing step.

It is important to consider that our algorithm has been trained on ImageCAS dataset which, even in the curated version, still suffers from data quality issues. Despite that, the proposed algorithm is capable of delivering solid results on unseen CGPS dataset, where it detected at least one of LAD and LCX branches in 93% of scans (comparing to 97.52% in ImageCAS). The results for CGPS data are slightly worse than for ImageCAS. It is worth noting that CGPS dataset consisted of 939 scans, potentially including a bit more variability in LCA structures than in case of ImageCAS test set of 162 scans. The proposed method is not free from heuristics that were tuned out on ImageCAS scans, and might not adapt perfectly to CGPS. Replacing heuristic-based approaches with more robust solutions could increase the algorithm's flexibility even more. It would be worthwhile to see how the algorithm performs when trained on a high quality data, not suffering from significant segmentation errors.

7.5.3 Comparison to state-of-the-art methods

For the final tracing algorithm we have chosen the one utilizing likelihood-sphere-based branch spawning, momentum and terminating conditions based on the artery likelihood. Our method requires only a starting point of the coronary artery to track the whole LCA. The employed spawning allows for efficient finding of tracing directions with limited heuristics and without unnecessary tracer's exploration. The momentum helps the tracer progress through the narrow fragments of the artery resulting in higher percentage of detected ground truth arteries. The terminating conditions based on the artery likelihood make for a robust indicator of when the tracer should be stopped.

The method achieves an OV of 0.812, $C_{G \rightarrow P}$ of 0.837 and AI of 0.097 evaluated on 162 images from ImageCAS dataset [2]. The achieved OV is slightly lower than the ones registered by the best state-of-the-art models (table 2.1). However, a lot of those algorithms were tested on a limited number of images, sometimes as few as 8. The AI of our solution is much lower than any of the state-of-the-art methods achieved. None of those methods reached AI lower than 0.21 mm, which makes the tracing accuracy of our algorithm stand out. Moreover, unlike in our project, many studies [11, 12] evaluate their result on the CAT08 dataset [63] which provides separate segmentation for LAD, RCA and LCX. Therefore, the metrics are calculated based solely on these parts of the coronary artery and do not reflect the capabilities of tracing the whole LCA.

Another strength of our approach lays in the terminating conditions. Some state-of-the-art methods [11, 13] terminate the tracer based on the uncertainty of the tracer network. This solution, even though often effective, may sometimes behave unexpectedly as the fact that the tracer network is uncertain does not necessarily mean that the branch is ending. Our terminating conditions are based on a tailor-made artery estimator tasked precisely with indicating when the tracer moves out of the artery.

Lastly, our algorithm proposes a novel branch spawning technique which is not reliant on explorative approach as is the case with other solutions [11, 12]. The tracing directions approximated by the model eliminate the need for exploring and later filtering potential branching candidates. Furthermore, our method does not rely on HU values unlike other approaches [13] when selecting the tracing path at bifurcation points. Instead, it utilizes the likelihood map built using the artery estimator. Using HU values to filter path candidates can mislead the tracer by choosing the paths going through the stenosis fragments as those disrupt the average HU calculation.

7.6 Phenotype clustering

On top of proposing the algorithm for tracing coronary artery, we employed a clustering method aimed at uncovering phenotypes of coronary artery tree morphology. With the use of hierarchical clustering applied on LM fragments, we found that the patients are best split between three phenotypes. Each phenotype has its own characteristics reflected in figures 6.25 and 6.26. The unique morphology of each type is linked to the risk of developing plaques.

Our results differ slightly from the ones presented by Wang et al., [17] where four clusters were found instead of three. The diverging number of clusters and slightly different profile of each phenotype might come from various factors such as the use of different dataset, inclusion of angle between coronary artery and the aorta, or optimal model selection method. Nevertheless, our findings also confirm that by making several relatively simple measurements based on the centerline positioning and artery radius one can identify patients with increased risk of suffering from atherosclerosis.

The scope of the analysis could also be expanded by including more parts of the coronary artery, or even the full tree.

The proposed solution, especially combined with the tracing algorithm, has the potential to automatically mark high risk patients based on the CCTA scan. When applied in a clinical setup, quicker identification of increased risk factors can support early intervention and prevent illness development [4]. The phenotype detection method can also be employed as a supporting tool in research of risk factors in cardiovascular diseases. Being able to label patients according to their phenotype could potentially shed a new light on which factors influence the likelihood of progression of CAD [8, 5].

In this study we have proposed an algorithm capable of tracing the left coronary trees based on CCTA scan and root point at the cross-section of LCA with the aorta. The method utilizes four neural network models combined into one final tracing algorithm.

The tracer CNN network is the main part responsible for guiding the tracer through the scan using a concept of direction quality score. The branch detector is tasked with identifying bifurcation points so that the tracer can be restarted to follow another branch. The radius estimator is a supporting model tasked with evaluating the radius of the coronary artery, while the artery estimator network is designed to predict the likelihood of a point laying inside the artery.

On top of all the models a concept of likelihood-sphere-based spawning has been introduced. This solution uses the artery and radius estimator models to scan the space around the branch point to create a likelihood map of correct tracing directions. The directions are approximated by utilizing the von Mises-Fisher modelling approach. Additionally, the algorithm employs a concept of momentum allowing for progressing through narrow fragments of the artery and robust terminating conditions based on artery likelihood.

The algorithm achieved highly satisfactory results recording an $OV = 0.812$, $C_{G \rightarrow P} = 0.837$ and $AI = 0.097$ mm. Our solution is capable of accurately detecting a substantial part of the whole coronary artery tree by using a single seed point at the root of the tree. The robustness of the results was proven with the deployment on CGPS dataset where the algorithm successfully detected at least one of LAD and LCX branches in 93% cases. Our approach stands out among the state-of-the-art solutions with the novel branch spawning approach based on the likelihood sphere, resilient terminating condition solution and momentum employed to detect arteries in difficult, narrow fragments.

On top of tracing algorithm, this study also proposes framework for deriving phenotypes based on the morphology of LMCA fragments. The phenotypes' characteristics are successfully linked to the risk factors of developing plaques.

Both of these solutions have the potential to be used either in clinical setup to aid early intervention and prevention of the CAD illnesses, or in research field as a complementary tools extending the scope of studies.

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